Cell injury

Lecture (2)

Cell injury

The term ‘cell injury’ is used to indicate a state in which the capacity for physiological adaptation is exceeded by excessive stimuli, or

when the cell is no longer capable to adapt without suffering some form of damage.
Cell injury may be reversible or irreversible.
Cell injury and cell death often result from exposure to toxic chemicals, infections and hypoxia (Figure 2.1).

Toxic chemicals

Chemical injury begins with the interaction between toxic chemicals and the plasma membrane.
Many classes of toxic chemicals are capable of inducing acute cell injury followed by death. These include anoxia and ischaemia and their chemical analogues such as:

potassium cyanide; chemical carcinogens, which form electrophiles that covalently bind to proteins in nucleic acids; oxidant chemicals, resulting in free radical formation and oxidant injury; activation of complement; and a variety of calcium ionophores.

Infections

Viruses induce cellular changes by two general mechanisms:

(1) cytolytic and cytopathic viruses cause various degrees of cellular injury and cell death,
(2) oncogenic viruses stimulate host cell to proliferate and may induce tumours.

Bacteria are relatively complex, single-celled creatures with a rigid wall and a thin, rubbery membrane surrounding the fluid inside the cell. They can reproduce on their own. Fossilized records show that bacteria have existed for about 3.5 billion years, and bacteria can survive in different environments,

including extreme heat and cold, radioactive waste and the human body.
Most bacteria are harmless, and some actually help by digesting food, destroying disease-causing microbes, fighting cancer cells and providing essential nutrients.

Hypoxia

Hypoxia is a deficiency of oxygen, which causes cell injury by reducing aerobic oxidative respiration.
Hypoxia is an extremely important and common cause of cell injury and cell death.
Causes of hypoxia include:

reduced blood flow, inadequate oxygenation of the blood due to cardiorespiratory failure and decreased oxygen-carrying capacity of the blood, as in anaemia or carbon monoxide poisoning (producing a stable carbon monoxyhaemoglobin that blocks oxygen carriage) or after severe blood loss.

Depending on the severity of the hypoxic state, cells may adapt, undergo injury or die.

Adaptation

Cells adapt to the environment to escape and protect themselves from injury. Cellular adaptations are common and a central part of many disease states.
The most significant adaptive states include atrophy, hypertrophy, hyperplasia and metaplasia.

Atrophy

Atrophy is a decrease or shrinkage in cell size caused by loss of subcellular organelles and substances (Figure 2.2). Atrophy can affect any, but it is most common in skeletal muscles, the heart, sex organs and the brain.
However, physiological atrophy occurs in some glands. For example, the thymus gland undergoes physiological atrophy during childhood.

Hypertrophy

This is an increase in the size of the cells, thus enlarging the size of the organ (Figure 2.3).
This can affect any cell but the cells of the heart, kidneys and skeletal muscles.

Hyperplasia

Hyperplasia is increased cell production in a normal tissue or organ (Figure 2.3). Hyperplasia may be a sign of abnormal or precancerous changes. This is called pathological hyperplasia.
Hyperplasia may be harmless and occur on a particular tissue.

An example of a normal hyperplastic response would be the growth and multiplication of milk-secreting glandular cells in the breast as a response to pregnancy, thus preparing for future breast-feeding.

Metaplasia

This is the reversible replacement of one differentiated cell type with another mature differentiated cell type.

The change from one type of cell to another may generally be a part of normal maturation process or caused by some sort of abnormal stimulus.

An example of metaplasia is the replacement of normal columnar ciliated epithelial cells of the bronchial lining by striated squamous epithelial cells.

Cells that die due to necrosis do not follow the apoptotic signal transduction pathway, but rather various receptors are activated that result in the loss of cell membrane integrity and an uncontrolled release of products of cell death into the intracellular space. This initiates an inflammatory response in the surrounding tissue.

Nearby phagocytes are prevented from locating and engulfing the dead cells.
The result is a build-up of dead tissue and cell debris at, or near, the site of the cell death.

Cell death

Cell death eventually leads to necrosis of the cell. It occurs when there is not enough blood flowing to the tissue, whether from injury, radiation, or chemicals.
Necrosis is not reversible.
One common type of necrosis is gangrene, which is often caused by damage from cold.

There are many types of necrosis, as it can affect many areas of the body, including bone, skin, organs and other tissues.

Apoptosis

Apoptosis is derived from the Greek words *apo*, meaning away from, and *ptosis*, meaning to fall.

The term ‘falling away from’ is derived from the fact that, during this type of prelethal change, the cells shrink and undergo marked blebbing at the periphery. The blebs then detach and float away.

It is sometimes referred to as programmed cell death and, indeed, the process of apoptosis follows a controlled, predictable routine. However, it is normal for many cells to die of apoptosis as the nervous system forms; it is part of constructing appropriate connections.

Apoptosis occurs in a variety of cell types following various types of toxic injury.
It is especially prominent in lymphocytes, where it is the predominant mechanism for turnover of lymphocyte clones.

The resulting fragments produce the basophilic bodies seen within macrophages in lymph nodes.

In other organs, apoptosis typically occurs in single cells, which are rapidly cleared away before and following death by phagocytosis of the fragments by adjacent parenchymal cells or by macrophages.

Apoptosis occurring in single cells with subsequent phagocytosis typically does not result in inflammation.

Prior to death, apoptotic cells show a very dense cytosol with normal or condensed mitochondria.
The endoplasmic reticulum (ER) is normal or only slightly dilated.

Inflammation

Lecture (3)

Inflammation

- Is the body's attempt at self-protection; the aim being to remove harmful stimuli, including damaged cells, irritants or pathogens, and begin the healing process.
  Inflammation can be defined clinically as the presence of swelling, redness and pain. Some diseases associated with inflammation include arthritis and neurological diseases.

- The signs and symptoms of inflammation are caused by four processes: (1) mast cell degranulation, (2) activation of plasma proteins, (3) the immune response and (4) heat. All these processes occur simultaneously to produce what is known as the inflammatory response.
- **First**, mast cell degranulation, is the release of granules containing serotonin and histamine from the mast cells into the tissues.
- These work with the other two processes below to provide the complete inflammatory signs and symptoms.

- **The second process** involves the activation of four plasma protein systems: complement (helps to orchestrate the inflammatory response); clotting (stops bleeding and repairs damage); kinin (involved in vascular permeability); and immunoglobulins (destroy bacteria), all of which work together to support the inflammatory process.
- This activates and assists inflammatory and immune processes, and also plays a major role in the destruction of bacteria.

- **The third process** is the movement of phagocytic cells to the area in order to phagocytose bacteria or any other non-self debris in the wound (Figure 3.2).
- **The fourth process, heat,** is a protective attempt by the organism to remove the injurious stimuli and initiate the healing process.
  Without this heating, wounds would never heal.

Physical and mechanical barriers

- These are part of the first-line defense against microorganisms. They include the skin and the epithelial cells of the viscera, genitourinary and respiratory tracts. The epithelial cells produce mucus to protect the lining of the tracts, some contain cilia to move the pathogens out and the temperature of the skin inhibits microorganisms from colonizing on the skin.

Biochemical barriers

- Epithelial surfaces also provide both physical and biochemical barriers against infection.
- Some of these substances include sweat, saliva and tears, which contain enzymes to destroy bacteria.
- Perspiration makes the skin pH slightly acidic, which is not a good environment for the bacteria to grow.

Acute and chronic inflammation

- Acute inflammation starts rapidly (rapid onset) and quickly becomes severe. Signs and symptoms are only present for a few days, but in some cases may persist for a few weeks.
  Some examples include acute bronchitis, appendicitis and sore throat.
- Chronic inflammation means long-term inflammation, which can last for several months and even years.
  Some examples include chronic asthma, chronic peptic ulcer and chronic sinusitis.

Tissue repair

- Wound healing, or cicatrization, is an intricate process in which the skin (or another organ-tissue) repairs itself after injury.
- In normal skin, the epidermis (outermost layer) and dermis (inner or deeper layer) exists in a steady-state equilibrium, forming a protective barrier against the external environment.

- Once the protective barrier is broken, the normal (physiological) process of wound healing is immediately set in motion.
- The classic model of wound healing is divided into three or four sequential, yet overlapping, phases: (1) haemostasis (not considered a phase by some), (2) inflammatory, (3) proliferative and (4) remodelling.
- Upon injury to the skin, a set of complex biochemical events take place in a closely orchestrated cascade to repair the damage.

- Within minutes post-injury, platelets (thrombocytes) aggregate at the injury site to form a fibrin clot. This clot acts to control active bleeding (**haemostasis**).
- The speed of wound healing can be impacted by many factors, including the bloodstream levels of hormones such as oxytocin.

- In the **inflammatory** phase, bacteria and debris are phagocytosed and removed, and factors are released that cause the migration and division of cells involved in the proliferative phase.

- The **proliferative** phase is characterized by angiogenesis, collagen deposition, granulation tissue formation, epithelialization, and wound contraction.
  - In **angiogenesis**, new blood vessels are formed by vascular endothelial cells.
  - In fibroplasia and **granulation tissue formation**, fibroblasts grow and form a new, provisional extracellular matrix (ECM) by excreting collagen and fibronectin.

- Concurrently, re-epithelialization of the epidermis occurs, in which epithelial cells proliferate and ‘crawl’ atop the wound bed, providing cover for the new tissue (Figure 3.3).

- In **contraction**, the wound is made smaller by the action of myofibroblasts, which establish a grip on the wound edges and contract themselves using a mechanism similar to that in smooth muscle cells. When the cells' roles are close to complete, unneeded cells undergo apoptosis.

- In the maturation and **remodelling** phase, collagen is remodelled and realigned along tension lines and cells that are no longer needed are removed by apoptosis.

- If the injury is minor then it is possible to return the injured tissues to their original structure and function through regeneration.
- However, if the injury is severe then regeneration is not possible and repair will not take place.
- Both regeneration and repair begin with phagocytosis, which includes fibrin from dissolved clots, microorganisms, erythrocytes and dead tissue.

- Three phases occur in repairing the wound. These are the migratory, proliferative and maturation phases.
  - In the **migratory phase**, the clot becomes a scab and epithelial cells migrate beneath the scab to bridge the wound.
  - During the **proliferative phase** there is extensive growth of epithelial cells beneath the scab, deposition of collagen fibres by fibroblasts and continued growth of blood vessels.

- In the **maturation phase**, the scab drops off as the epidermis returns to normal thickness.
- In the dermis the, collagen fibres become more structured, fibroblasts decrease in number, and blood vessels are restored to their normal function.

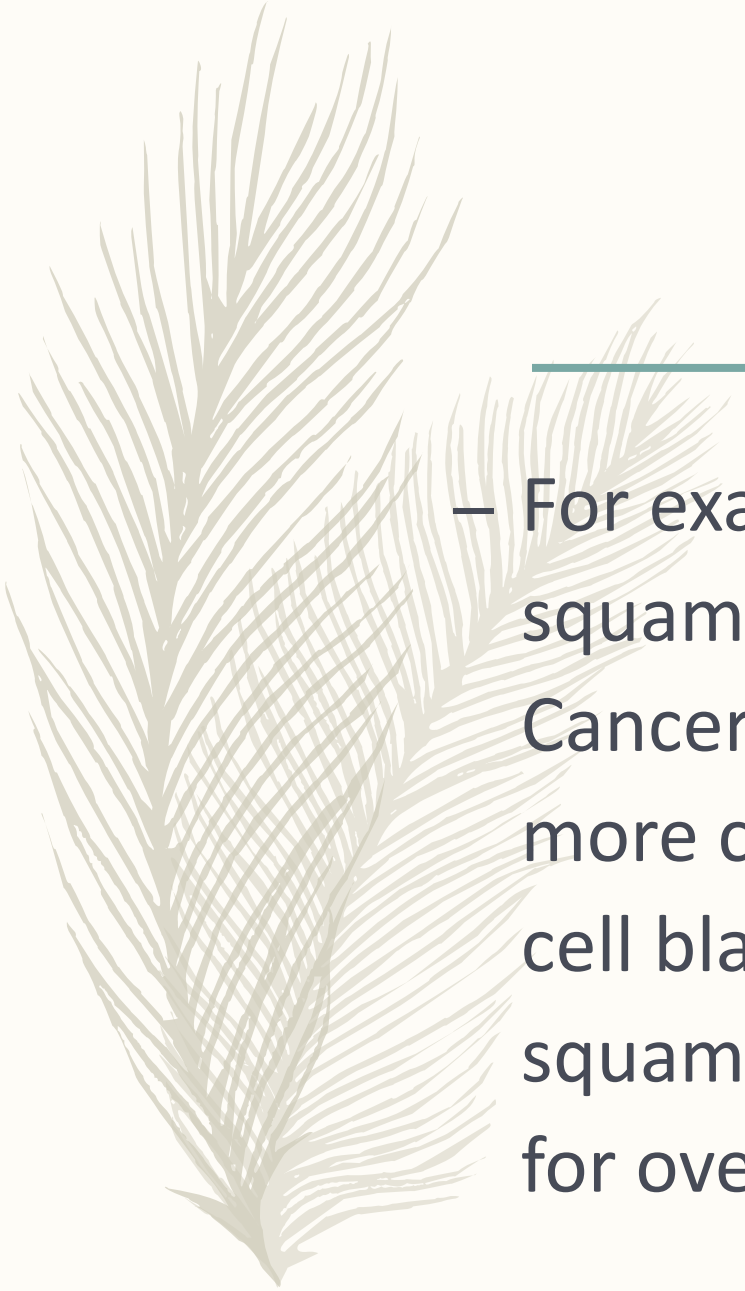
Cancer

Lecture (4)

What is cancer

– Cancer is a disease of the cells in the body. There are many different types of cell in the body, and many different types of cancer which arise from different types of cell. What all types of cancer have in common is that the cancer cells are abnormal and multiply out of control. If not treated, the tumour can cause problems in one or more of the following ways.

- Cancer can develop from almost any type of cell in the body. So there is usually more than one type of cancer that can develop in any one part of the body. Often, though, one type of cancer will be much more common in a particular organ (Figure 4.1).



– For example, there are cells called transitional cells, squamous cells and adenomatous cells in the bladder. Cancer can develop in any of these cells, but is much more common in the transitional cells. Transitional cell bladder cancer is much more common than squamous cell cancer or adenocarcinoma, accounting for over nine out of ten cases of bladder cancer.

How many types of cancer are there?

- It is estimated that there are more than 200 different types of cancer.

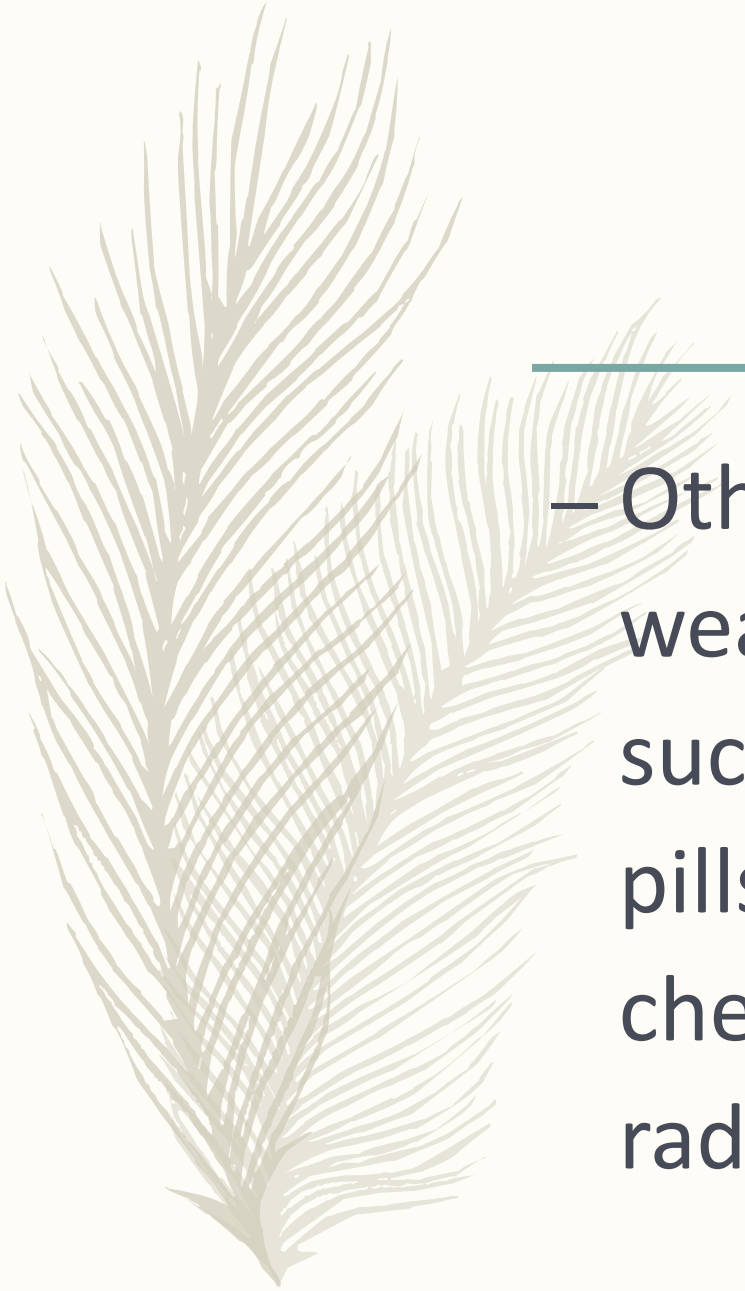
  Cancer can develop in any organ of the body. There are over 60 different organs in the body where a cancer can develop.

  Each organ is made up of several different types of cells.

  For example, the skin has several layers, including connective tissue, often containing gland cells. Underneath that there is often a layer of muscle tissue and so on. Each type of tissue is made up of specific types of cells.

Causes of cancer

– Causes of cancers can be classified into two groups: **first**, there are genotoxic carcinogens which affect the DNA, causing mutations; and **second**, promoter substances, which may cause hormonal imbalance, altered immunity or long-term tissue damage, which can lead to the development of cancer.



- Other risk factors include viruses as they weaken the immune system, hormones, such as the ones found in contraceptive pills, which can lead to breast cancer, chemical agents such as arsenic, and radiation.

Genetic factors

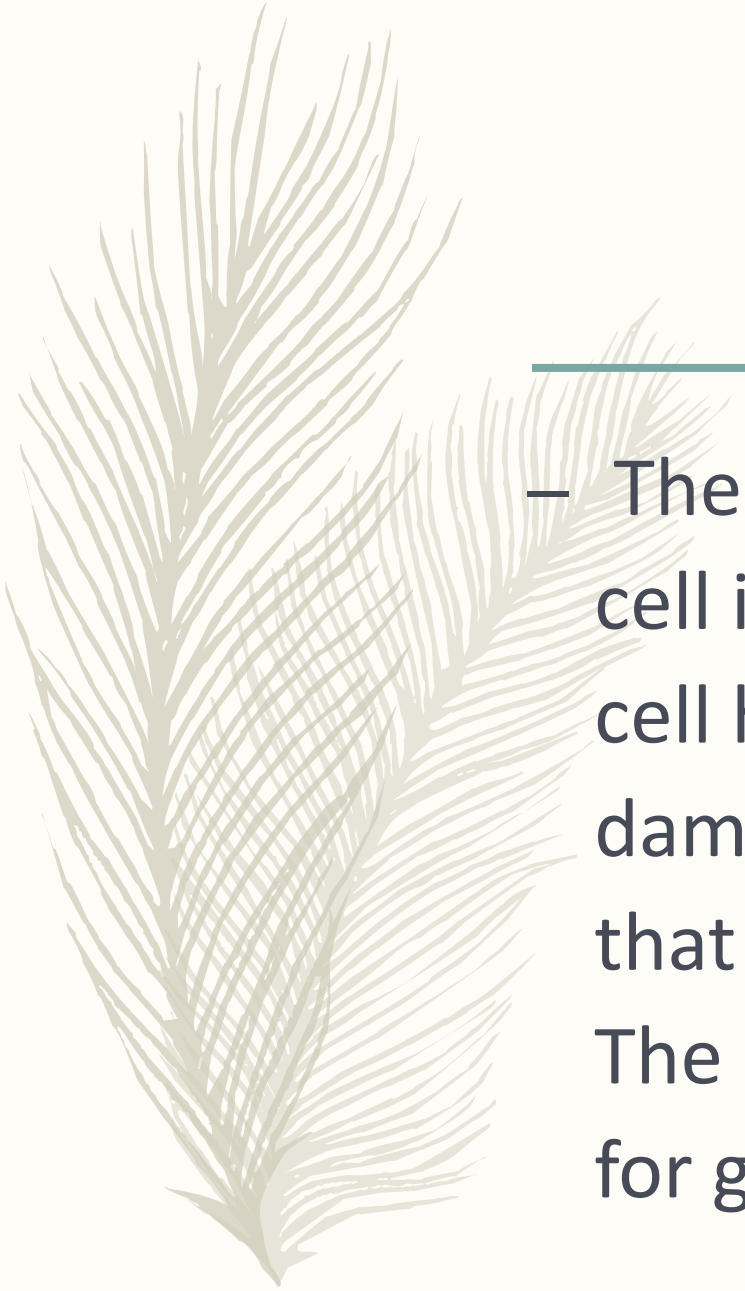
- There need to be a number of genetic mutations within a cell before it becomes cancerous. Sometimes a person is born with one of these mutations already. This doesn’t mean they will definitely get cancer. But with one mutation from the start, it makes it more likely statistically that they will develop cancer during their lifetime. Doctors call this genetic predisposition.

- The BRCA1 and BRCA2 breast cancer genes are examples of genetic predisposition. Women who carry one of these faulty genes have a higher chance of developing breast cancer than women who do not.

- The BRCA genes are good examples for another reason. Most women with breast cancer do not have a mutated BRCA1 or BRCA 2 gene. Less than 3% of all breast cancers are due to these genes.
  So although women with one of these genes are individually more likely to get breast cancer, most breast cancer is not caused by a high-risk inherited gene fault.

Age

- Most types of cancer become more common as a person get older. This is because the changes that make a cell become cancerous in the first place take a long time to develop.
There have to be a number of changes to the genes within a cell before it turns into a cancer cell.



- These changes can happen by accident when the cell is dividing, or they can happen because the cell has been damaged by carcinogens and the damage is then passed on to future cells when that cell divides.

  The longer a person lives the more time there is for genetic mistakes to happen in the cells.

Viruses

- Viruses can help to cause some cancers. But this does not mean that these cancers can be caught like an infection.
  What happens is that the virus can cause genetic changes in cells that make them more likely to become cancerous.

Bacterial infection

- Bacterial infections have not been thought of as cancer-causing agents in the past. But studies have shown that people who have Helicobacter pylori (H pylori) infection of their stomach develop inflammation of the stomach lining, which increases the risk of stomach cancer.
  Helicobacter pylori infection can be treated with a combination of antibiotics.

Immune system

- People who have problems with their immune systems are more likely to get some types of cancer.
  This group includes people who have had organ transplant and HIV.

Classification/grading/staging

- Diagnosis consists of naming the tumour (classification), describing its aggressiveness (grading) and reporting how far it has spread (staging). The stage of a cancer describes its size and whether it has spread beyond the area of the body where it started. The grade of a cancer depends on what the cells look like under the microscope.
See Figure 4.2.



- Stage 0: is where it started (in situ) and not spreading.
- Stage I: the tumour is less than 2 cm and is not spreading.
- Stage II: the tumour is 2–5 cm with or without lymph node involvement (lymph nodes are part of the lymphatic system). It has not spread.



- Stage III: the tumour is more than 5 cm, or any size, but fixed either to the chest wall, muscle or skin, or has spread to lymph nodes above the collarbone.
- Stage IV: the tumour is any size – it may affect the lymph nodes but has definitely spread to other parts of the body.

Lecture 5

Cardiogenic shock

- Cardiogenic shock occurs when there is diminished cardiac output and evidence of tissue hypoxia in the presence of adequate intravascular volume. The leading cause of death in acute myocardial infarction (MI) is cardiogenic shock.
  Cardiogenic shock is a physiological state in which inadequate tissue perfusion results from cardiac dysfunction, usually systolic. It is a major and often fatal complication of a number of acute and chronic disorders, occurring most often after acute MI.

Pathophysiology

- When there is failure of the pump action of the heart, resulting in reduced cardiac output, cardiogenic shock has occurred. Cardiogenic shock can be defined as the presence of the following (regardless of adequate left ventricular filling pressure): sustained hypotension (systolic blood pressure less than 90 mm Hg for more than 30 minutes) and tissue hypoperfusion (cold peripheries, or oliguria less than 30 ml/hour, or both).

- A low cardiac output state secondary to extensive left ventricular infarction, development of a mechanical defect (e.g. ventricular septal defect or papillary muscle rupture) or right ventricular infarction are conditions that can result in the acute deterioration of cardiac function. This can lead to cardiogenic shock, including MI or myocardial ischaemia, acute myocarditis, sustained arrhythmia, severe valvular dysfunction and decompensation of end-stage cardiomyopathy as a result of multiple aetiologies.

- Those who develop cardiogenic shock from acute MI regularly have evidence of progressive myocardial necrosis with infarct extension (myocardial pathology). Decreased coronary perfusion pressure and increased myocardial oxygen demand play a role in the vicious cycle that leads to cardiogenic shock.
  There is usually multi-vessel coronary artery disease, with limited coronary blood flow reserve.

- Tissue hypoperfusion, with resulting cellular hypoxia, results in anaerobic glycolysis, the accumulation of lactic acid, and intracellular acidosis (cellular pathology).
  If ischaemia is severe and protracted, myocardial cellular injury becomes irreversible, leading to myonecrosis, which includes mitochondrial swelling, the accumulation of denatured proteins and chromatin and lysosomal breakdown. These acts induce fracture of the mitochondria, nuclear envelopes and plasma membranes.

- Large areas of myocardium that are dysfunctional but still viable can contribute to cardiogenic shock in patients with MI (reversible myocardial dysfunction).
  This potentially reversible dysfunction is described as myocardial stunning or as hibernating myocardium.
  Myocardial stunning represents post-ischaemic dysfunction persisting despite the restoration of normal blood flow.

- As a result of decreased contractility, the person develops elevated left and right ventricular filling pressures and low cardiac output. Mixed venous oxygen saturation falls due to the increased tissue oxygen extraction, because of the low cardiac output. Combined with the intrapulmonary shunting that is often present, this contributes to substantial arterial oxygen desaturation.

- As a mass of left ventricular myocardium becomes ischaemic, failing to pump effectively, stroke volume and cardiac output are decreased. Myocardial ischaemia is further exacerbated by compromised myocardial perfusion due to hypotension and tachycardia, resulting in systemic effects.

- Systemic perfusion is compromised by reduced cardiac output, with tissue hypoperfusion intensifying anaerobic metabolism and instigating the formation of lactic acid, which further worsens the systolic performance of the myocardium.

- Compensatory mechanisms are activated, including sympathetic stimulation, increasing heart rate and cardiac contractility, causing renal fluid retention and as such augmenting the left ventricular preload.
  Elevated heart rate and contractility increases demand for myocardial oxygen, worsening myocardial ischaemia.

- Pulmonary venous congestion and hypoxaemia occurs, amplifying myocardial afterload, which additionally impairs cardiac performance. Excessive myocardial oxygen demand, with simultaneous inadequate myocardial perfusion, worsens myocardial ischaemia, a vicious cycle is initiated, which if uninterrupted leads to death.

- Myocardial ischaemia decreases myocardial compliance, elevating left ventricular filling pressure at a given end-diastolic volume (diastolic dysfunction), leading to pulmonary congestion and congestive heart failure.

- Shock state, regardless of the cause is described as a syndrome initiated by acute systemic hypoperfusion, leading to tissue hypoxia and vital organ dysfunction.
  The heart plays a central role in propagating shock. Reduced coronary perfusion leads to worsening cardiac dysfunction with a cycle of self-perpetuating progression of global hypoperfusion ensuing.
  Renal compensation for reduced perfusion causes diminished glomerular filtration, resulting in oliguria and subsequent renal failure.

Signs and symptoms

- These are hypotension, absence of hypovolaemia and clinical signs of poor tissue perfusion, including oliguria, cyanosis, cool extremities and confusion.
  Physical examination includes cool, ashen or cyanotic skin and extremities are mottled. Peripheral pulses are rapid and faint, and may be irregular if arrhythmias are present.

- Jugular venous distention and crackles in the lungs may be present; peripheral oedema can also be present. Heart sounds are distant and third and fourth heart sounds may be present. Pulse pressure may be low, and patients are usually tachycardic. There is hypoperfusion, altered mental status and decreased urine output.

Investigations

- **First-line investigations** can help to determine the underlying cause of cardiogenic shock. See Table 5.1.
- Blood pressure measurement
- Urea and electrolytes, and creatinine
- Liver function tests and full blood count FTs
- Cardiac enzymes, including troponins
- Arterial blood gases

Cont. … Investigations

- Brain natriuretic peptide (BNP)
- Electrocardiogram
- Chest X-ray
- CT pulmonary angiography (CTPA) or ventilation/perfusion lung scan
- Echocardiography
- Coronary catheterization (angiogram).

Management

- Management is based on an individual assessment of each patient. **Cardiogenic shock treatment** focuses on repairing the damage to heart muscle and other organs caused by lack of oxygen. **Emergency life support** is a necessary treatment for most people who have cardiogenic shock.

- Symptom relief may be needed, for example opiate analgesia.
  Treat any electrolyte abnormalities.
  Treat any cardiac arrhythmias.

  See Table 5.2.

Anaphylactic shock

LECTURE (6)

Anaphylaxis

is a serious, potentially life-threatening allergic response that is marked by swelling, hives, hypotension and dilated blood vessels.
In severe cases shock may occur and if anaphylactic shock is not treated immediately, it can result in death.

Anaphylaxis occurs when the immune system develops a specific antibody (called immunoglobulin E, or IgE) that drives an inappropriate or exaggerated reaction toward a substance that is usually harmless, for example food.
The body may not react upon initial exposure but may produce antibodies with later exposures.
When exposed to the substance later, the binding of the allergen to antibodies can lead to the presence of a large amount of histamine (a protein) being produced, causing the symptoms identified above.

Anaphylactic shock

Anaphylaxis can occur within seconds or minutes of exposure to something the person is allergic to, for example the venom from a bee sting.
The immune system releases a flood of chemicals during anaphylaxis that can cause shock.
Common triggers of anaphylaxis include certain foods, some medications, insect venom and latex.
Anaphylaxis requires immediate emergency treatment; if the condition is not treated right away, unconsciousness or even death can occur.

Pathophysiology

The number of those who suffer severe systemic allergic reactions is growing; however, as anaphylaxis is not always recognized, the incidence may be under-reported.

An anaphylactic reaction happens when an allergen reacts with specific IgE antibodies on mast cells and basophils (type 1 hypersensitivity reaction).

This triggers the rapid release of stored histamine and the rapid synthesis of newly formed mediators.
As a result these cause capillary leakage, mucosal oedema and eventually shock and asphyxia.
Anaphylactic reactions can vary in severity and rate of progression; they may progress rapidly (over a few minutes) or sometimes in a biphasic manner. Rarely, manifestations may be delayed by a few hours (this adds to diagnostic difficulty), or continue for more than 24 hours.

Anaphylactoid reactions are not IgE mediated but cause similar mast cell activation.
A significant number of cases of anaphylaxis are idiopathic.
Table 6.1 identifies the most common triggers of anaphylaxis.

Signs and symptoms

There is usually (but not always) a previous history of sensitivity to an allergen, or recent history of exposure to a new drug.
Primarily, people often develop skin symptoms, including generalized itching, urticaria and erythema, rhinitis, conjunctivitis and angio-oedema.
If the airway is becoming involved signs may include itching of the palate or external auditory meatus, dyspnoea, stridor (laryngeal oedema) and bronchospasm.

General symptoms comprise palpitations and tachycardia, nausea, vomiting and abdominal pain, a fainting feeling, with a sense of impending doom; ultimately the person will collapse and lose consciousness.
If there is airway swelling, stridor, dysnpoea, bronchospasm, cyanosis, hypotension, tachycardia and reduced capillary filling then these are suggestive of an approaching severe reaction. See Figure 6.1.

Investigations

Anaphylaxis is diagnosed based on its symptoms. Those with a history of allergic reactions may be at greater risk for developing a severe reaction in the future. **Skin testing** can help confirm the substances that cause severe allergic reactions. However, this type of test may not be recommended if there is a reason to suspect that the person will have an anaphylactic reaction to the substance.

Serum mast-cell tryptase can be measured in order to clarify diagnosis where ambiguity exists. Tryptase is the preferred marker demonstrating mast-cell degranulation. Levels of serum tryptase, which is a mast-cell specific protease, are highest at one hour post an anaphylactic reaction, staying elevated for about six hours. Elevated serum tryptase levels imply massive mast-cell degranulation (occurring in anaphylaxis) or a condition such as mastocytosis.
Not all cases of anaphylaxis result in a rise in tryptase.

National guidance exists concerning the identification and management of anaphylaxis. The essential principles of treatment are the same for all age groups.
Diagnosis of anaphylaxis is a clinical one, beginning with a brief, directed history, as treatment must be provided quickly. This history includes questions about a previous history of atopy or anaphylaxis and exposure to new foods, medications and insect bites or stings.

Management

Rapid assessment:

Airway: look for and relieve airway obstruction; call for help early if there are signs of obstruction. Remove any traces of allergen remaining (e.g. nut fragments caught in teeth, with a mouthwash, bee stings, without compressing any attached venom sacs).

Breathing: look for and treat bronchospasm and signs of respiratory distress.

Circulation: colour, pulse and BP.

Disability: assess whether responding or unconscious.

Exposure: assess skin with adequate exposure, but avoid excess heat loss.

- High-flow oxygen is given using a mask with an oxygen reservoir (greater than 10 litres per minute to prevent reservoir collapse).
- The person is placed flat: Epinephrine injection is the only rapidly effective treatment for anaphylaxis; epinephrine rapidly reverses anaphylactic symptoms. It is typically given through an automatic injection device. The most common and most effective injection site is the thigh.

- Large-bore intravenous cannulae are inserted and a rapid fluid challenge is administered.
- Assessment of central venous pressure is required.
- Chlorphenamine and hydrocortisone are given after initial resuscitation.
- Pulse oximetry, electrocardiograph and blood pressure are monitored.

Those patients prone to anaphylaxis should be provided with one or more auto-injector kits and should be instructed on their use.
The patient should be advised to carry the auto-injector kit on their person at all times.

Hypovolaemic Shock

LECTURE (7)

- Hypovolaemic shock occurs when the volume in the circulatory system is too depleted to allow adequate circulation to vital organs and tissues of the body. Resuscitation aims to correct the hypovolaemia and hypoperfusion of vital organs, such as the brain and kidneys, prior to irreversible damage occurring.

Hypovolaemic shock

- Hypovolaemic shock is a medical or surgical condition; rapid fluid loss results in multiple organ failure as a result of inadequate circulating volume, causing inadequate perfusion.
  Hypovolaemic shock is very often due to rapid blood loss (sometimes referred to as haemorrhagic shock).

- Two common causes of haemorrhagic shock are acute, external blood loss secondary to penetrating trauma and severe gastrointestinal bleeding.
  It can also be caused by a significant acute internal bleed into the thoracic and abdominal cavities.

- Hypovolaemic shock can result from significant fluid loss, other than blood loss, for example fluid loss due to severe gastroenteritis and extensive burns.

Pathophysiology

- Without fluid and blood resuscitation and/or correction of the underlying pathology causing the haemorrhage, cardiac perfusion eventually diminishes, and multiple organ failure soon follows.

- There are a number of causes of hypovolaemic shock, including loss of blood; this can be revealed or occult. Trauma can lead to visible bleeding or rupture of an internal organ such as the spleen or liver.

- A fractured femur can bleed approximately half a litre and a fractured pelvis will lose about one litre of blood (note these volumes vary according to age/weight).
  The amount of haematemesis indicates the degree of bleeding.
  Bleeding as a result of ectopic pregnancy is also occult, with little or no vaginal loss.

- Blood loss and leakage of plasma occurs in burns.
  Severe loss of water and salt occurs with strenuous exercise in a hot environment, poor fluid intake, loss from diarrhoea and vomiting, and inappropriate diuresis.

- Other causes of shock include, poor cardiac output as is the case in massive myocardial infarction, ruptured aortic valve or massive pulmonary embolism, septic shock and acute pancreatitis, leading to a fall in vascular resistance and high transection of the spine or a high spinal anaesthetic.

- A healthy adult can endure the loss of half a litre from a circulation of approximately five litres without adverse effect; larger volumes and rapid loss, however, result in progressively greater problems.
  Risk is related to the degree of hypovolaemia as well as the speed of correction.

- The risk of morbidity and mortality grows as age increases. Unfavorable pathology in the cardiovascular, respiratory and renal systems can increase risk.

- The human body has an ability to respond to acute haemorrhage by activating the following major physiological systems:
  the haematological, cardiovascular, renal and neuroendocrine systems.

- The haematological system responds to a severe acute blood loss by triggering the coagulation cascade and contracting the bleeding vessels (releasing local thromboxane A2 release).

- Platelets are activated and form an immature clot on the source of bleeding.
  The damaged vessel exposes collagen; this subsequently causes fibrin deposition and stabilization of the clot.

- Initially the cardiovascular system responds to hypovolaemic shock by increasing the heart rate, increasing myocardial contractility and constricting peripheral blood vessels.
  This occurs secondary to an increased release of norepinephrine and decreased baseline vagal tone (the baroreceptors in the carotid arch, aortic arch, left atrium and pulmonary vessels regulate this).

- The cardiovascular system redistributes blood to the brain, heart and kidneys and away from skin, muscle and gastrointestinal tract.

- The renal system responds to haemorrhagic shock, stimulating an increase in the secretion of renin from the juxtaglomerular apparatus. Angiotensinogen is converted to angiotensin I by renin, which is then converted to angiotensin II by the lungs and liver.

- Angiotensin II has two important effects, both of these help to reverse haemorrhagic shock, vasoconstriction of arteriolar smooth muscle and stimulation of aldosterone secretion by the adrenal cortex.

- The neuroendocrine system responds to haemorrhagic shock by causing an increase in circulating antidiuretic hormone (ADH).
  This is released from the posterior pituitary gland in response to a decrease in blood pressure (detected by baroreceprors) and a decrease in the sodium concentration (detected by osmoreceptors).
- ADH indirectly leads to an increased reabsorption of water and sodium chloride by the distal tubule, the collecting ducts, and the loop of Henle.

- The pathophysiology of hypovolaemic shock is complex; there are many intricate mechanisms that are effective in maintaining vital organ perfusion in severe blood loss. See Figure 7.1.

Signs and symptoms

- The person might feel cold, unwell, anxious, faint and short of breath.
  There may be faintness on standing or even on sitting up, related to postural hypotension.
  There may be symptoms related to the cause of the hypovolaemia, for example as pain from a haemorrhaging ulcer, dissecting aneurysm, ruptured ectopic pregnancy, trauma or burns.

- Nausea and vomiting can be related to gut ischaemia.
  The patient may look pale and sweaty, with tachypnoea.
  The periphery may be cold due to poor perfusion; capillary refill time may be prolonged.

- There may be tachycardia and hypotension or postural hypotension. Tachycardia is often a later feature than the cold periphery, and hypotension occurs still later.
  Late features include confusion or even coma.
  Physiologically there are three stages of hypovolaemic shock. See (Table 7.1).

Investigations

- These include full blood count, urea and electrolytes, liver function tests, coagulation screen and arterial blood gas analysis.
- Ultrasound is used for differentiating hypovolaemic from cardiogenic shock; the vena cava can be assessed for adequate filling.

- Echocardiogram may demonstrate pump failure.
- Central venous pressure monitoring may be of value where there is evidence of shock.

Management

- Rapid volume repletion is required in those with hypovolaemic shock. Delayed therapy can lead to ischaemic injury and potentially to irreversible shock and multi-organ system failure.
  Table 7.2 outlines the fundamentals of management.

Table 7.2 The fundamentals of management

General measures	• Oxygen should be given • Venous access is required early • Resuscitation usually begins with crystalloid, such as normal saline or Hartmann's solution. Albumin may play a role in those requiring substantial amounts of crystalloid to maintain a mean arterial pressure in the setting of septic shock. In haemorrhage, blood should be given as soon as possible • A central venous pressure line may be required
Pharmacological	• Vasodilator therapy may be required. It is used more for septic shock. If hypovolaemia occurs then circulating blood volume should be increased • Analgesia must be given by the intravenous route; any other route will be ineffective. Pain increases metabolic rate and so aggravates tissue ischaemia
Surgical	• Surgery may be needed if bleeding continues to stem the flow. It is normal to resuscitate first, reducing the risk, as induction of anaesthesia can lead to collapse of a fragile circulation • In trauma, first aid measures may help stem blood loss. In gunshot wounds look for not just the entrance but also the exit wound. The latter may be significantly larger than the former. If a high-powered weapon was used, primary closure of the wound should not be attempted

Human immunodeficiency virus (HIV)

Human immunodeficiency virus

- HIV was initially named lymphadenopathy-associated virus (LAV). It was the third identified in a family of human T-lymphotropic viruses (HTLV-III), finally renamed as HIV-I, and causes acquired immune deficiency syndrome (AIDS) related disease in most parts of the world.

- Previously other acronyms were used to describe an illness experienced by a predominately gay male population in North America, for example GRID, gay-related immune deficiency. HIV-II was identified in 1985 in people with AIDS with West African connections and is uncommon outside that region.

- Human immunodeficiency virus (HIV) is a retrovirus that was first identified in the early 1980s.
  A person who has been infected with HIV can remain asymptomatic for eight years or more.
  Once this virus begins to destroy helper T cells it progresses very quickly taking 2–5 years if untreated.

- When the T-cell count drops to below 200 cells/μL or when there are opportunistic infections, such as toxoplasmosis, cytomegalovirus, or Pneumocystis (carinii) jiroveci pneumonia (PCP), cancer or acquired immunodeficiency syndrome (AIDS) related dementia the person is diagnosed as having AIDS.

- It is important to note that infection with HIV is not AIDS. There are cases where the person with HIV (without treatment) has been known to survive for longer than twelve years with no signs of developing AIDS.
HIV infection, however, does mean that the individual is infectious to others with or without the symptoms associated with AIDS.

- HIV is a blood-borne virus typically transmitted via sexual intercourse, sharing of intravenous drug paraphernalia and mother to-child transmission, occurring during the birthing process or during breast-feeding.

Pathophysiology

- HIV gains entry to the cells by attaching to the CD4 receptor and a co-receptor via its envelope glycoproteins.
  It encodes the enzyme reverse transcriptase, permitting a DNA copy to be made from viral RNA.

- Once integrated into the cellular DNA the provirus exists in the nucleus of infected cells, where it can remain dormant for lengthy periods of time.
  It can also become transcriptionally active using the machinery of the host cell to replicate.

- Viral RNA is then modified singly or in multiples, making numerous structural and regulatory and accessory proteins.
Viral proteases further process proteins; mature viral particles are formed as budding of the virus occurs through the host cell membrane (see Figure 9.1).

- There is a high level of viral replication in the blood within a few weeks of infection, and an associated decline in CD4 T cells occurs.
  An immune response to HIV develops, curtailing viral replication, causing a decrease in viral load and a return of CD4 T- cell numbers close to normal levels.

- Immune control is said to be dependent on killer T cells and neutralizing antibodies.
  It has been suggested that the host's initial response to HIV infection is critical and that this is genetically governed.

- CD4 cells migrate to the lymphoid tissue, where the virus replicates and then infects new CD4-positive cells.

  As the infection progresses, depletion or impaired function of CD4 cells predisposes to the development of immune dysfunction.

Signs and symptoms

- There may be flu-like symptoms, a glandular fever type illness, fever, malaise, myalgia, pharyngitis, headaches, diarrhoea, neuralgia or neuropathy, lymphadenopathy and/or a maculopapular rash.
  Rarely, meningoencephalitis occurs.
  Acute infection may be asymptomatic.

- HIV seroconversion illness includes oral candidiasis, recurrent shingles, leukopenia, or central nervous system signs (e.g. seizures, dementia).

Investigations

- HIV infection should be considered in any patient with unusual or recurrent serious infections without another cause, particularly in those with risk factors for HIV infection. Table 9.1 lists those groups of people who should be offered a routine HIV test.

Table 9.1 People who should be offered a routine HIV test

- Patients where HIV enters the differential diagnosis
- Those diagnosed with a sexually-transmitted infection
- Sexual partners of people known to be HIV positive
- Men who have sexual contact with other men
- Female sexual contacts of men who have sexual contact with men
- Those who have ever injected drugs
- People from countries of a high prevalence of HIV – greater than 1%
- People who have had sexual contact abroad or in the UK with individuals from countries with high HIV prevalence
- The following in relation to current guidance:
  - blood donors
  - dialysis patients
  - organ transplant donors and patients

- A high-sensitivity enzyme-linked immunoabsorbent assay (ELISA) should be used for screening; a positive result should be followed with confirmatory testing, such as Western blot assays or similar specific assay.
HIV-II should be tested for in patients from an HIV-II endemic area or those with indeterminate results on HIV-I Western blot testing.

- In August 2013, rapid HIV testing for the simultaneous detection of HIV-I p24 antigen as well as antibodies to both HIV-I and HIV-II in human serum, plasma and venous or fingerstick whole blood specimens was approved.

Management

- Screening for HIV infection is essential, as people infected can remain asymptomatic for years as the infection progresses. Serological tests are the most important studies in the evaluation for HIV infection.

- CD4 counts vary; the reference range for CD4 counts is 500– 2000 cells/µL. After seroconversion, CD4 counts decrease (on average 700/µL), declining over time.
The rate of progression to AIDS and death is related to the viral load. Those with viral loads greater than 30,000/µL are 18.5 times more likely to die of AIDS than those with undetectable viral loads. (see Figure 9.2).

The treatment of HIV disease depends on the stage of the disease and any associated opportunistic infections.
The goal of treatment is to prevent the immune system from deteriorating so that opportunistic infections become more likely.

- Highly active antiretroviral therapy (HAART) is the principal method for preventing immune deterioration.
  Prophylaxis for specific opportunistic infections is indicated in specific cases.

- As there is no cure for HIV (currently) the prevention of HIV is through health education and the promotion of safer sex, which is essential; the aim is to reduce exposure to HIV contaminated body fluid.

NON-HODGKIN LYMPHOMA

NON-HODGKIN LYMPHOMA

- Non-Hodgkin lymphomas (NHLs) are the fifth most common cancer in the UK, with approximately 11,500 people diagnosed with it annually.
  There are several different types of NHL.
  Some of them will grow very slowly and treatment may not be required for months or years, other types have a tendency to grow quickly and treatment is required soon after a diagnosis is made.

- Non-Hodgkin's lymphoma is associated with ageing. As a person ages there are increased chances of developing the condition.
  At diagnosis the average age is around 65.
  Risk is higher in white people than black and Asian people.

PATHOPHYSIOLOGY

- Non-Hodgkin lymphomas (NHLs) are a heterogeneous group of tumours that originate from lymphoid tissues, primarily of lymph nodes (a type of cancer of the lymphatic system).
(see Figures 10.1 and 10.2.)

- They are lymphoproliferative malignancies with differing patterns of behaviour and responses to treatment; the prognosis depends on the histological type, stage and treatment. With NHLs there is a much greater tendency for the malignancy to disseminate to extranodal sites than is the case with Hodgkin's lymphoma.

- NHL is associated with an altered or defective immune system; the incidence of the illness is higher in those people who are receiving immunosuppressive medications for organ transplant, people with HIV and those people with certain viral conditions, for example Epstein-Barr virus.

- Some risk factors include having a family member with the condition (family history), being male, being white, having an autoimmune disease such as rheumatoid arthritis and having been exposed to radiotherapy.

- NHL may result from chromosomal translocations, infections, environmental factors, immunodeficiency states and chronic inflammation.

  The malignant biochemical characteristic in NHL is due to abnormal gene mutations occurring during lymphocyte production, maturation or action.

- The transformations lead to growth advantage and expansion of a monoclonal population of malignant lymphocytes.
  The type of lymphoma (B cell or T cell) depends on the stage of lymphocyte production, maturation or action at which mutation occurs.

B-CELL LYMPHOMA

- The process of B-cell production is still undergoing much research. It is thought that B cells originate and mature in the bone marrow (the central lymphoid tissue compartment).
  They may leave bone marrow to perform their function in lymph nodes and extranodal tissues (e.g. peripheral lymphoid tissue compartment).

- The abnormal mutations can occur at the stage of early precursors, leading to corresponding subtypes of acute lymphoblastic leukaemia. Immature B cells, mature antigen-naive B cells and mature antigen-activated B cells may transform to various types of NHL, such as Burkitt's lymphoma, diffuse large B-cell lymphoma and mantle cell lymphoma.

T-CELL LYMPHOMA

- The exact cause of most T-cell lymphomas is unknown. However, it is likely that genetic changes that take place during antigen processing and presentation result in an aggressively dividing cell population that may give rise to T-cell lymphoma.

- In contrast to B cells, T cells migrate early into the thymus, where they develop into mature cells.

  The abnormal mutations can happen at different stages of T-cell development and can lead to different malignant phenotypes.

SIGNS AND SYMPTOMS

- The clinical manifestations vary with such factors as the location of the lymphomatous process, speed of tumour growth and the function of the organ compromised or displaced by the malignancy.
  Clinical manifestations can include:

LOW-GRADE LYMPHOMAS:

- Painless, slowly progressive, peripheral lymphadenopathy.
- Primary extranodal involvement.
- Fatigue, weakness, fever, night sweats, weight loss.
- Cytopenia • Splenomegaly • Hepatomegaly.

INTERMEDIATE AND HIGH-GRADE LYMPHOMAS:

- Rapidly growing, bulky lymphadenopathy.
- Systemic symptoms and extranodal involvement.
- Hepatomegaly • Splenomegaly • Testicular mass
- Skin lesions: associated with cutaneous T-cell lymphoma.
- Burkitt's lymphoma: presents with a large abdominal mass and symptoms of bowel obstruction.
- Pulmonary involvement, superior vena obstruction.

INVESTIGATIONS

- Full blood count • Renal function tests and electrolytes • Serology: HIV, HTLV-1, hepatitis C
- Chest X-ray
- Lymph node excision core biopsy is for the diagnosis of follicular lymphoma; histology for assessment of the tumour grade.

- CT scan of neck, chest, abdomen and pelvis
- Bone scans
- Whole body positron emission tomography (PET) scan.
- Scrotal ultrasound.
- Lumbar puncture.
- Other investigations may be indicated, depending on the clinical presentation.

MANAGEMENT

- Treatment options vary because of the mixed nature of NHLs: the type and grade of NHL, stage of NHL and the person's general health. Options include: watchful waiting, single-agent or multi-agent chemotherapy and regional or extended radiotherapy.

- Surgery in treatment is limited but may be useful in certain situations.
- Those with severe neutropenia should be given antibiotic prophylaxis with chemotherapy.
- Monoclonal antibodies (rituximab, a biological therapy) may be given with chemotherapy.

Anaemia

Lecture (13)

Overview of anatomy and physiology

- Blood is composed of plasma and blood cells. The blood cells are erythrocytes (red blood cells), leukocytes (white blood cells) and thrombocytes (platelets). These are suspended in the plasma with other particulate matter (Figure 16.1).

- Plasma is a clear straw- coloured fluid that makes up more than half the volume of blood. The red blood cells and proteins contribute to the viscosity of the blood, which ranges from 3.5 to 5.5 compared with 1.000 for water. The more red blood cells and plasma proteins in blood, the higher the viscosity and the slower the flow of blood.

- Plasma is the liquid part of the blood and is composed of water (91%), protein (8%) (albumin, globulin, prothrombin and fibrinogen), salts (0.9%) (sodium chloride, sodium bicarbonate and others) and the remaining 0.1% is made up of organic materials,

- for example fats, glucose, urea, uric acid, cholesterol and amino acids. These substances give plasma greater density and viscosity than water.

- Plasma proteins make up 7% of the plasma and these proteins stay in the blood vessel as they are too large to diffuse through capillaries and are responsible for creating the osmotic pressure of blood.
  When plasma proteins are lost in patients who suffer from burns, fluid moves into tissues, causing oedema by a process called osmosis.

Pathophysiology

- Anaemia is a condition where the amount of haemoglobin in the blood is below the normal level, or there are fewer red blood cells than normal. There are several different types of anaemia and each one has a different cause, although iron deficiency anaemia is the most common type. Other forms include vitamin B12 deficiency, and aplastic anaemia.

Iron deficiency anaemia

- Iron deficiency anaemia occurs when there isn't enough iron in the body thus reducing the amount of red blood cells (Figure 16.2). Iron is found in meat, dried fruit and some vegetables. Iron is used by the body to make haemoglobin, which helps store and carry oxygen in red blood cells.

- Excess iron is stored in the liver and muscle cells, and is readily available for the production of red blood cells. Some inflammatory disorders such as Crohn's disease will affect the absorption of iron from the gastrointestinal tract, affecting the synthesis of red bloods cells.

Vitamin B12/folate deficiency anaemia

- Pernicious anaemia is the most common cause of vitamin B12 deficiency in the UK.
  Pernicious anaemia is an autoimmune condition affecting the stomach.
  An autoimmune condition means the immune system (the body's natural defence system that protects against illness and infection) attacks the body's healthy cells.

- Pernicious anaemia causes the immune system to attack the cells in the stomach that produce the intrinsic factor.
  This means the body cannot absorb vitamin B12, which causes a deficiency.

- Vitamin B12 and folate work together to help the body to produce red blood cells. Vitamin B12 helps to keep the nervous system (brain, nerves and spinal cord) healthy; folate is important for pregnant women because it reduces the risk of birth defects in unborn babies.

Aplastic anaemia

- Aplastic anaemia is a rare, potentially life-threatening failure of haemopoiesis characterized by pancytopenia and hypocellular bone marrow. Aplastic anaemia is defined as pancytopenia with a hypocellular bone marrow in the absence of an abnormal infiltrate and with no increase in reticulin.

- Most cases are acquired and immune-mediated but there are also inherited forms. Environmental triggers include drugs, viruses and toxins, but most cases are idiopathic.
  Drugs that cause aplastic anaemia may also be related to benzene. The antibiotic, chloramphenicol and the anti-inflammatory, phenylbutazone are two examples.

What are the symptoms of aplastic anaemia?

- The person complains of increasing tiredness, weakness and shortness of breath.
  Bleeding, bruising and blood spots may be noticed.
  Sore throats and other infections are noticeable.
  A high temperature with shivering attacks is an important symptom that demands immediate medical attention.

Signs and symptoms

- Symptoms of each type of anaemia vary depending on the underlying cause of the condition. However, there are some general symptoms associated with all types of anaemia. These include: fatigue, breathlessness, lethargy, faintness, tinnitus, headache, hair loss, constipation and loss of appetite.

Management

- The management of anaemia depends on which type the patient has. Iron deficiency anaemia is treated with oral iron medication and less commonly parenterally.
  The common medications used are ferrous gluconate and ferrous sulphate.

- Prior to the administration of iron medication, nurses should be aware of any drugs that may interact with iron. In syrup form the iron suspension can stain the teeth and patients should be made aware of this.

- Constipation, black stools and dysphagia are some of the side effects of iron tablets and therefore the patient may need laxatives to relieve the discomfort of constipation.
  Folic acid tablets are used to restore folate levels, which usually need to be taken for four months.

- Information on food which is rich in iron should be given to patients.
  These include red meat, leafy vegetables and cereals.
- Polyphenols found in tea and coffee can impair iron absorption and therefore patients should be advised to limit these beverages until the level of iron is restored in the body.

- Patients who find it difficult to get enough vitamin B12 in their diets, such as vegans, may need vitamin B12 for life.

Deep vein thrombosis

Lecture (14)

Overview of anatomy and physiology

- A DVT is a blood clot in a vein.
  Blood clots in veins most often occur in the legs, but can occur elsewhere in the body, including the arms (Figure 17.1).
- The most common cause of a blood clot developing in a vein is immobility.
- A complication can occur in some cases where part of the blood clot breaks off and travels to the lung (pulmonary embolus).

- Blood flows throughout the body tissues in blood vessels, via bulk flow (i.e. all constituents together and in one direction).
  The cardiovascular system is a closed-loop system, such that blood is pumped out of the heart through one set of vessels (arteries) and then returns to the heart in another (veins).

- The pulmonary circulation is composed of the right heart pump and the lungs. The systemic circulation includes the left heart pump, which supplies blood to the systemic organs.
  The right and left heart pumps function in a series arrangement; thus both will circulate an identical volume of blood in a given minute (cardiac output: litres per minute).

- The aorta's four principal divisions are the ascending aorta (begins at the aortic valve where, close by, the two coronary artery branches have their origin), arch of the aorta, thoracic aorta and abdominal aorta.
The smallest of the arteries eventually branch into arterioles. They, in turn, branch into an extremely large number of the smallest diameter vessels, the capillaries (with an estimated 10 billion in the average human body).
Next, blood exits the capillaries and begins its return to the heart via the venules.

- Capillaries, which are the smallest and most numerous blood vessels in the human body (ranging from 5 to 10 micrometres in diameter and numbering around 10 billion), are also the thinnest walled vessels; an inner diameter of 5 μm is just wide enough for an erythrocyte to squeeze through (Figure 17.2).

Pathophysiology

- The pathophysiology of DVT was described by Virchow in 1846 as coming from a triad of possible changes in the venous system.
  The triad includes changes in the vessel wall (injury), changes in the pattern of blood flow (venous stasis) and changes in the constituency of blood (hypercoagulability).

- These physiological changes can occur as a result of pathology, therapies and treatments.
  Injury to the vessel wall may occur from trauma, surgery or invasive treatments.
  Patients on bed rest have venous stasis, which changes blood flow thorough the venous system, and patients on new medications may have changes in blood coagulability.

- A thrombus can develop in the superficial or deep veins of the legs. The blood flow is sluggish in the affected vessels and clotting cascade takes place. Platelets aggregate at the site of injury to the vessel wall or where there is venous stasis.

- Platelet aggregation occurs because platelets are exposed to collagen (a protein in the connective tissue which is found in the inner surface of the blood vessel). When platelets come into contact with the exposed collagen, they release adenosine diphosphate and thromboxane.

- These substances make the surface of the platelets (Figure 17.3) become sticky and as they adhere to each other a platelet plug is formed. Other cells, such as the red blood cells, are trapped in the fibrin meshwork and the thrombus grows.

Signs and symptoms

- Most of the time, DVT may not present with any symptoms.
  The possible signs and symptoms may include unexplained pyrexia and dull aching pain in the affected limb.

- Localized pain or tenderness within a calf or thigh muscle is a possible symptom of DVT, particularly if there is swelling in the calf below the knee or redness or warmth in that area with a heavy or aching sensation.

Management

- Anticoagulant medicines prevent a blood clot from getting bigger. They can also help stop part of the blood clot from breaking off and becoming lodged in another part of the bloodstream (an embolism).

- Although they are often referred to as ‘blood-thinning’ medicines, anticoagulants do not actually thin the blood.
  They alter chemicals within it, which prevents clots forming so easily.

- Two different types of anticoagulants are used to treat DVT: heparin and warfarin.
  Heparin is usually prescribed first, because it works immediately to prevent further clotting.
  After this initial treatment the patient may also need to take warfarin to prevent another blood clot forming.

- Warfarin is taken as a tablet.
  It may be taken after an initial heparin treatment to prevent further blood clots occurring.
  It is usually recommended that warfarin is taken for 3–6 months.
  In some cases, warfarin may need to be taken for longer, even for life.

- Compression stockings help prevent calf pain and swelling and lower the risk of ulcers developing after having a DVT.
  They can also help prevent post-thrombotic syndrome – damage to the tissue of the calf caused by the increase in venous pressure that occurs when a vein is blocked (by a clot) and blood is diverted to the outer veins.

Leukaemia

Lecture (15)

Overview of anatomy and physiology

- Leukaemia is a cancer of cells in the bone marrow (the cells which develop into blood cells). Leukaemia is a malignant disorder where there is an abnormal or excessive proliferation of immature white blood cells. In the UK, leukaemia is the twelfth most common cancer in adults, affecting more men than women.

- All of the blood cells in the body are produced by bone marrow.
  Bone marrow is a spongy material found inside the bones. It is important because it produces special cells called stem cells. Stem cells are very useful because they have the ability to create other specialized cells that carry out important functions.

- The large flat bones, such as the pelvis and breastbone (sternum), contain the most bone marrow.
  To make blood cells constantly a healthy bone marrow and nutrients from the diet are needed, including iron and certain vitamins.

- Usually the bone marrow produces stem cells that are allowed to mature into ‘adult’ blood cells (Figure 18.1).
  Stem cells are primitive (immature) cells.
  There are two main types in the bone marrow: myeloid and lymphoid stem cells.

- These derive from even more primitive common ‘pluripotent’ stem cells (Figure 18.2). Stem cells constantly divide and produce new cells.
  Some new cells remain as stem cells and others go through a series of maturing stages (‘precursor’ or ‘blast’ cells) before forming into mature blood cells.
  Mature blood cells are released from the bone marrow into the bloodstream.

- However, in cases of acute leukaemia, the affected bone marrow begins to release a large number of immature white blood cells that are known as blast cells.
  There are two principal types and they are acute and chronic leukaemia.
  Each of these is further subdivided into: acute ALL and AML and chronic CLL and CML.

Risk factors

- Men are more likely to develop CML, CLL and AML than women.
  The risk of most leukaemias, with the exception of ALL, typically increases with age.
  Patients who have had certain types of chemotherapy and radiation therapy for other cancers have an increased risk of developing certain types of leukaemia.

- People exposed to very high levels of radiation, such as survivors of a nuclear reactor accident, have an increased risk of developing leukaemia.
  Exposure to certain chemicals, such as benzene – which is found in gasoline and is used by the chemical industry – is also linked to increased risk of some kinds of leukaemia.

Pathophysiology Acute leukaemia

- Acute leukaemia is more aggressive and develops rapidly. It is more common in the younger age group; the symptoms develop quickly and if untreated become life threatening.
  Leukaemic cells are immature and poorly differentiated; they proliferate rapidly, have a long lifespan and do not function normally.

- AML is overproduction of immature myeloid white blood cells and ALL is the overproduction of immature lymphocytes called lymphoblasts.
  In acute leukaemia, the cells reproduce very quickly, and do not become mature enough to carry out their role in the immune system.

- Acute leukaemia is characterized by a rapid increase in the number of immature blood cells.
  Crowding due to such cells makes the bone marrow unable to produce healthy blood cells.
  Acute forms of leukaemia are the most common forms of leukaemia in children.

Chronic Leukaemia

- Chronic leukaemia is an uncommon type of cancer. About 600 people in the UK are diagnosed with chronic myeloid leukaemia each year.
- Chronic myeloid leukaemia can affect people of any age, but it is more common in people aged 40–60.
- There is no evidence that it runs in families.

- Chronic leukaemia is characterized by the excessive build-up of relatively mature, but still abnormal, white blood cells.
  Typically taking months or years to progress, the cells are produced at a much higher rate than normal, resulting in many abnormal white blood cells.

- Whereas acute leukaemia must be treated immediately, chronic forms are sometimes monitored for some time before treatment to ensure maximum effectiveness of therapy.
  Chronic leukaemia mostly occurs in older people, but can theoretically occur in any age group.

Signs and symptoms

- The symptoms vary depending on the exact type of leukaemia; however, the common presenting symptoms include: tiredness, breathlessness, abnormal bleeding from the gums, abdominal pain due to enlarged spleen or liver, weight loss and swollen lymph glands in the groin, neck and under the arms.

Management

- Patients suffering from leukaemia will need an accurate and full assessment of pain level, activity tolerance, vital signs, nutrition, signs of bleeding or infection, in order to plan high-quality care.

- Avoid situations in which trauma may occur, such as shaving with straight-edge razor, ambulating after taking medication that may cause orthostasis, or using sharp utensils.
- Advise the patient on the importance of maintaining hydration and taking good nutrition.
- Advise the patient to take frequent rests when feeling tired as fatigue is a problem for patients with leukaemia.

- Adhere to local trust protocols in the care of a patient with leukaemia. Nurses must wash their hands before and after attending to the patient and discourage unnecessary visits by relatives and friends.

Pharmacological and non-pharmacological treatments

- Some medications that may be prescribed include cytotoxic drugs and opoiod drugs to control pain.
  Other interventions include radiotherapy and bone marrow transplant.

- Some patients may be treated with ATRA (All-trans retinoic acid).
  ATRA works by changing the immature blast cells into mature healthy blood cells, and can reduce symptoms very quickly.

Thrombocytopenia

LECTURE (16)

Overview of anatomy and physiology

- Thrombocytopenia is the term for a reduced platelet (thrombocyte) count. It happens when platelets are lost from the circulation faster than they can be replaced from the bone marrow where they are made.

What are platelets?

- Platelets are tiny cells (Figure 19.1) that circulate in the blood and whose function is to take part in the clotting process.
  Inside each platelet are many granules, containing compounds that enhance the ability of platelets to stick to each other and also to the surface of a damaged blood vessel wall.

- The platelet count in the circulating blood is normally between 150 and 400 million per millilitre of blood. Newborn babies have a slightly lower level, but are normally within the adult range by three months of age.

- Platelets are produced in the bone marrow, as are the red cells and most of the white blood cells. Platelets are produced from very large bone marrow cells called megakaryocytes.

- As megakaryocytes develop into giant cells, they undergo a process of fragmentation that results in the release of over 1000 platelets per megakaryocyte. The dominant hormone controlling megakaryocyte development is thrombopoietin.

Function of platelets

- Platelets are essential in the formation of blood clots to prevent haemorrhage – bleeding from a ruptured blood vessel. An adequate number of normally functioning platelets is also needed to prevent leakage of red blood cells from apparently uninjured vessels.

- In the event of bleeding, muscles in the vessel wall contract and reduce blood flow. The platelets then stick to each other (aggregation) and hold on to the vessel wall (primary haemostasis).
  The coagulation factors are then activated, resulting in normally liquid blood becoming an insoluble clot.

Pathophysiology

- Three distinct types of thrombocytopenia have been identified and they are ITP, TTP and HUS.
- **ITP** (Idiopathic Thrombocytopenic Purpura) is a disease in which antibodies form and destroy the body's platelets. As the destruction is believed to be caused by the body's immune system,

- it is therefore classified as an autoimmune disorder.
  Although the bone marrow increases the synthesis of platelets, it cannot keep up with the demand.

- Acute ITP is more common in children, while chronic ITP is more common in adults. Acute ITP may require no treatment, especially if the platelet count does not fall too low and there is little bleeding.
  It usually improves spontaneously and, in children at least, rarely comes back.

- Chronic ITP may not need treatment if the platelet level doesn't pose a significant risk of bleeding.
  Any such assessment should take account of the lifestyle, such as participation in contact sports or manual work.

- Platelets become coated with antibodies as a result of autoimmune response mediated by B lymphocytes. Although the platelets function normally, the spleen identifies them as foreign protein and destroys them.

- **TTP** (Thrombotic thrombocytopenic purpura) is a rare disease in which small blood clots form suddenly throughout the body. The numerous amounts of blood clots result in a high level of platelet usage in clotting, which leads to a reduction in platelets.

- **HUS** (Hemolytic-uremic syndrome) is a rare disorder related to TTP in which the number of platelets decreases and there is reduction in the number of red blood cells. HUS can also occur with intestinal infections with *Escherichia coli* and with the use of some drugs, such as cyclosporine.

Drug-induced thrombocytopenia

- Almost all medications can cause bad allergic reactions in sensitive people, but these reactions are rare.
  Drugs can also cause serious reactions with blood platelets. In these reactions, drugs stick to the platelet surface,

- and this combination of the drug bound to the platelet can be recognized by the body as a foreign substance and the body then makes an antibody to the drug-coated platelets, and all platelets can be destroyed.

Signs and symptoms

- Thrombocytopenia symptoms may include: easy or excessive bruising, superficial bleeding into the skin that appears as a rash of petechiae, usually on the lower legs, prolonged bleeding from cuts,

- spontaneous bleeding from gums or nose, blood in urine or stools, unusually heavy menstrual flows and profuse bleeding during surgery or after dental work.

Management

- As a result of a low level of platelets, the patient is at risk of bleeding, especially from the gums. Early identification of bleeding is important in order to prevent blood loss.

- Nursing interventions include prevention, early identification and management of thrombocytopenia and its complications. Take steps to minimize the patient's risk of injury, such as preventing falls, reducing the bleeding risk and preventing infection.

- Be aware that thrombocytopenia increases the risk of infection in patients receiving corticosteroids. Steroid-related skin changes, bruising and bleeding under the skin increase the risk of skin breakdown.

- Microemboli and altered coagulation raise the risk of ischaemia of the extremities and altered organ function. To minimize these problems, ensure continuity of care and evaluate skin and organ function.

- Monitor the patient's fluid intake and output to evaluate hydration status and check stools for occult blood.
  Remember that in thrombocytopenia, increased bleeding may lead to hypovolaemia, which can affect sensory perception and set the stage for falls and renal dysfunction.

- If the patient requires isolation due to immunosuppressants or infection, undertake a psychosocial assessment to evaluate psychosocial needs.

- Advise the patient and family how to avoid injury.
  Determine the types of assistance the patient may need with activities of daily living.
  Advise patients and family members how to recognize signs and symptoms of infection and ischaemia.

HEART FAILURE

Lecture (17)

OVERVIEW OF ANATOMY AND PHYSIOLOGY

Heart failure means that the heart is not pumping blood around the body as well as it used to.

The most common reason is that heart muscle has been damaged, for example after a heart attack.

Common causes of heart failure include myocardial infarction and other forms of ischemic heart disease, hypertension, valvular heart disease and cardiomyopathy.

The electrical impulse travels first through the walls of the atria, which contract to pump blood into the ventricles.

The impulse then travels through to walls of the ventricles, which contract to pump blood into the arteries.

PATHOPHYSIOLOGY

Heart failure is caused by any condition which reduces the efficiency of the myocardium, or heart muscle, through damage or overloading.
As such, it can be caused by as diverse an array of conditions as myocardial infarction (in which the heart muscle is starved of oxygen and dies),

hypertension (which increases the force of contraction needed to pump blood) and amyloidosis (in which protein is deposited in the heart muscle, causing it to stiffen).
Over time these increases in workload will produce changes to the heart itself.

Reduced contractility, or force of contraction, is due to overloading of the ventricle.
In a healthy heart, increased filling of the ventricle results in increased contractility (by the Frank-Starling law of the heart) and thus a rise in cardiac output.

In heart failure this mechanism fails, as the ventricle is loaded with blood to the point where heart muscle contraction becomes less efficient. This is due to reduced ability to cross-link actin and myosin filaments in over-stretched heart muscle.

The general effect is one of reduced cardiac output and increased strain on the heart.
This increases the risk of cardiac arrest (specifically due to ventricular dysrhythmias), and reduces blood supply to the rest of the body.

In chronic disease the reduced cardiac output causes a number of changes in the rest of the body, some of which are physiological compensations.

The physiological compensations include destimulation of baroreceptors in the carotid sinus and aortic arch, which link to the solitary nucleus. This centre in the brain increases sympathetic activity, releasing catecholamines into the bloodstream, resulting in vasoconstriction.

Reduced perfusion (blood flow) to the kidneys stimulates the release of renin – an enzyme which catalyses the production of the potent vasopressor angiotensin.

Angiotensin and its metabolites cause further vasocontriction, and stimulate increased secretion of the steroid aldosterone from the adrenal glands. This promotes salt and fluid retention at the kidneys, also increasing the blood volume.

Ankle swelling related to heart failure is usually better in the morning and gets worse later in the day, although this is the case for most causes of ankle swelling.

Other symptoms include: coughing, wheezing, weight gain or loss, extreme tiredness, feeling sick, tachycardia and pulmonary oedema.

MANAGEMENT

In order to provide high-quality care, healthcare professionals need to undertake a full and accurate assessment and devise a care plan for all the actual problems identified. Vital signs are monitored hourly until they are stable.

Early detection in changes in vital signs and prompt treatment may save the patient's life.

Advise the patient who is a smoker to stop smoking, and provide support with smoking cessation.

Advise patients regarding good nutrition and provide help for obese patients to reduce their weight.

Patients should be nursed in an upright position in bed supported by pillows unless contraindicated.

Alcohol can act as a negative inotrope, and increase blood pressure and the risk of arrhythmias.

Advise patients with severe CCF, particularly in conjunction with hyponatraemia, to restrict their fluid intake sensibly.

Take care to avoid excessive dehydration – particularly in elderly patients on high-dose diuretics.

PHARMACOLOGICAL TREATMENT

Heart failure could be treated with a combination of drugs. These drugs help people with systolic heart failure live longer and feel better.

ACE inhibitors are a type of vasodilator, a drug that widens blood vessels to lower blood pressure, improve blood flow and decrease the workload on the heart.

Digoxin (Lanoxin), also referred to as digitalis, increases the strength of the heart muscle contractions. It also tends to slow the heartbeat. Digoxin reduces heart failure symptoms in systolic heart failure.

Beta blockers are used to reduce the risk of some abnormal heart rhythms and lessen the chance of dying unexpectedly.
Beta blockers may reduce signs and symptoms of heart failure, improve heart function, and help patients live longer.

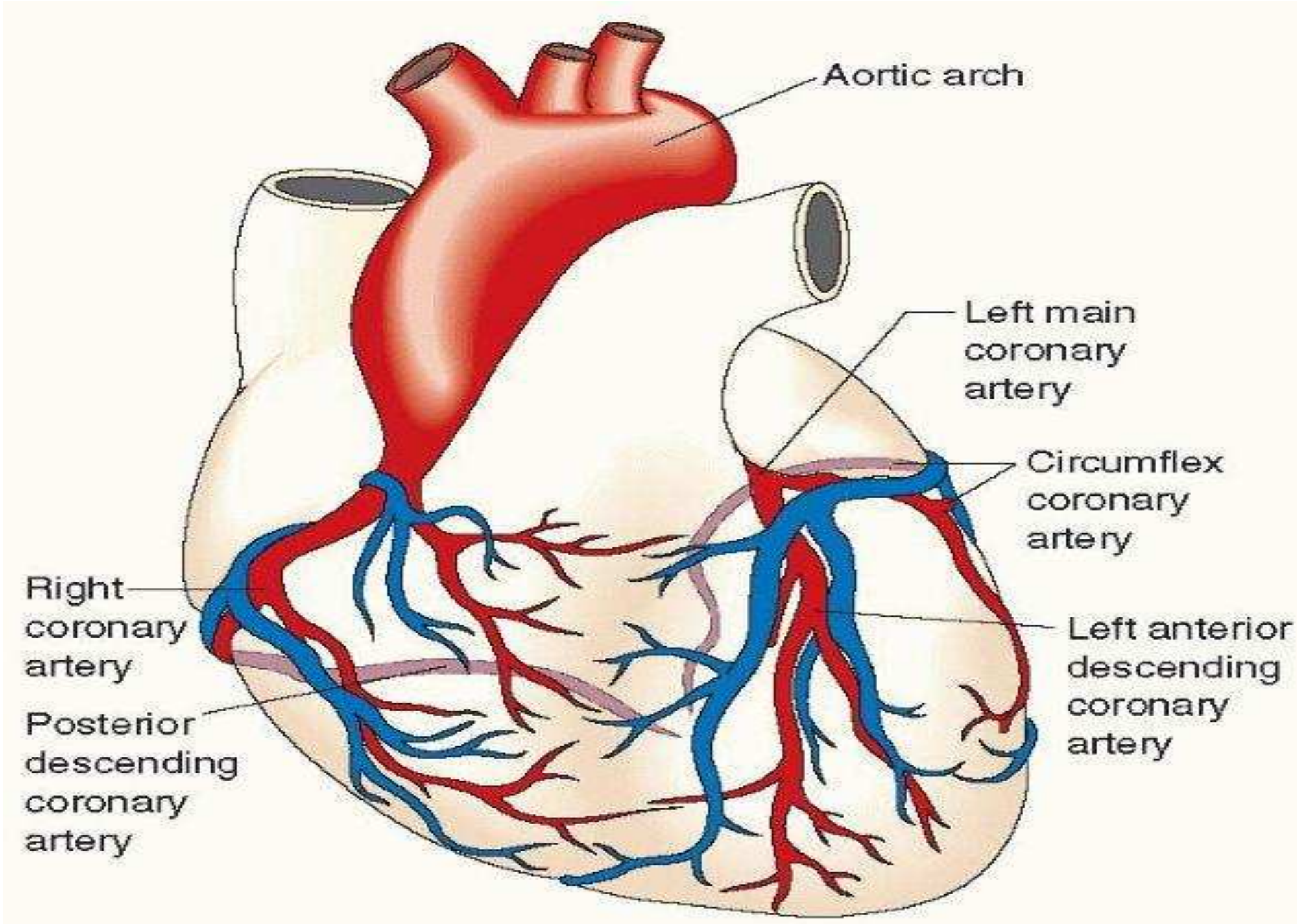
Other medications include diuretics, aldosterone antagonists and inotropes.

Myocardial infarction

Lecture (18)

Overview of anatomy and physiology

- The heart is mainly made of special muscle (myocardium). The heart pumps blood into arteries (blood vessels) which take the blood to every part of the body.
  Like any other muscle, the heart muscle needs a good blood supply.
  The coronary arteries take blood to the heart muscle (Figure 21.1).


Aortic arch
Left main coronary artery
Circumflex coronary artery
Right coronary artery
Left anterior descending coronary artery
Posterior descending coronary artery

- The main coronary arteries branch off from the aorta (the large artery which takes oxygen-rich blood from the heart chambers to the body.)
  The main coronary arteries divide into smaller branches which take blood to all parts of the heart muscle.

Pathophysiology

- The most common cause of an MI is a blood clot (thrombosis) that forms inside a coronary artery, or one of its branches. This blocks the blood flow to a part of the heart. Blood clots do not usually form in normal arteries. However, a clot may form if there is some atheroma within the lining of the artery.

- Atheroma is like fatty patches or plaques that develop within the inside lining of arteries (Figure 21.2). (This is similar to water pipes that get furred up.)
  Plaques of atheroma may gradually form over a number of years in one or more places in the coronary arteries.
  Each plaque has an outer firm shell with a soft inner fatty core.

- In an MI (heart attack), a coronary artery or one of its smaller branches is suddenly blocked, which results in the damage of the myocardium (Figure 21.3).
  The part of the heart muscle supplied by this artery loses its blood (and oxygen) supply if the vessel is blocked.

- The area of infarct occurs in the distribution of the occluded vessel. Left main coronary artery occlusion generally results in a large anterolateral infarct, whereas occlusion of the left anterior descending coronary artery causes necrosis limited to the anterior wall.
  Where the infarct has taken place, a collagen scar forms in its place and the damaged muscle does not contract efficiently.

- Collagen is a bundle of inelastic fibres that do not stretch or contract effectively. Damaged heart tissue conducts electrical signals much more slowly than normal heart tissue, which could result in inefficient contraction of the myocardium.

Signs and symptoms

- A person having an acute myocardial infarction usually has sudden chest pain that is felt behind the breast bone and sometimes travels to the left arm or the left side of the neck (Figure 21.4). Additionally, the person may have shortness of breath, sweating, nausea, vomiting, abnormal heartbeats and anxiety.

- Rapid, irregular pulse, hypotension and dyspnoea (shortness of breath) may all present as symptoms.
  The anxiety is often described as a ‘sense of impending doom’.
  Dyspnoea occurs when the damage to the heart limits the output of the left ventricle, causing left ventricular failure and consequent pulmonary oedema.

- Women experience fewer of these symptoms than men, but usually have shortness of breath, weakness, a feeling of indigestion, and fatigue.
  In many cases, in some estimates as high as 64%, the person does not have chest pain or other symptoms. These are called ‘silent’ myocardial infarctions.

Management

- Patients suffering from an acute myocardial infarction will be maintained on bed rest to minimize cardiac work.
  Reassurance will be necessary for both the patient and relatives as they will be anxious.

- Pain relief is paramount, current guidelines recommend the use of morphine (5–10 mg) or diamorphine (2.5–5 mg) titrated to pain. A popular method of titrating pain relief is to make a 10 mg dose of morphine in 10 ml of water for injection and administer the morphine in 1 ml (1 mg) increments until pain relief is achieved.

- The benefit of opiate pain relief is that it also acts to relieve anxiety in the patient. However, as opiates are associated with nausea and vomiting it is advisable to administer an antiemetic at the same time. The patient is also attached to continuous cardiac monitoring as the risks of cardiac arrhythmias and even cardiac arrest are high.

- **Some experts** still recommend the use of sublingual nitrate spray in the patient suffering from myocardial infarction.
  If intravenous morphine is ineffective then the recommendation is for the use of intravenous nitrates or beta blocker drugs.
  Before the use of nitrates or beta blockers it is necessary to measure the patient's blood pressure, as both nitrates and beta blockers can significantly reduce the blood pressure.

Pharmacological and non-pharmacological treatment

- Anticoagulant, such as heparin, is commenced to minimize the risk of a thrombus developing. Drugs to dissolve clots, such as reteplase or streptokinase, are administered within two hours of developing MI to limit tissue damage. A continuous ECG monitoring is carried out to detect abnormal cardiac rhythms to take prompt action.

- In some patients, an emergency coronary angioplasty may be required to increase blood flow to the coronary arteries. This involves insertion of a catheter into the obstructed coronary artery under local anaesthesia. The balloon in the catheter is then inflated for 15 seconds to two or three minutes, which dilates the artery.

Angina Pectoris

LECTURE (19)

Overview of anatomy and physiology

Angina is a pain that comes from the heart.
It is usually caused by narrowing of the coronary (heart) arteries (Figure 23.1).
Usual treatment includes a statin medicine to lower cholesterol level, low-dose aspirin to help prevent a heart attack and a beta blocker medicine to help protect the heart and to prevent angina pains.

- The heart is mainly made of special muscle. The heart pumps blood into arteries which take the blood to every part of the body. Like any other muscle, the heart muscle needs a good blood supply. The coronary arteries take blood to the heart muscle.

- They are the first arteries to branch off the aorta. This is the large artery taking blood from the heart to the rest of the body.
  In angina, one or more of the coronary arteries is usually narrowed. This causes a reduced blood supply to a part, or parts, of the heart muscle.

Pathophysiology

- Angina pain closely resembles the signs and symptoms of MI; thus it is vital that carers are able to differentiate between the two conditions, as the treatment differs in both cases.
  If a patient has angina pain, this is usually relived by vasodilators, for example Nitroglycerin or GTN (glyceryl trinitrate), but the angina pain is rarely fatal.

- Angina results from a blockage in the coronary arteries, giving diminished blood supply to the affected part of the heart muscle.
  At rest the blood supply may be sufficient to provide nutrients and oxygen to the heart muscle; however, during activity,

- for example walking or running, the heart rate increases, which puts more effort on the heart.
  During exertion, if the blood flow to the heart muscle is inadequate, the oxygen supply is also diminished, leading to severe pain.
  The patient may present with pallor, dyspnoea, cyanosis, diaphoresis and tachycardia.

Types of angina

- **Stable angina** This is the most common type of angina, and pain is usually brought about by physical exertion that results in the heart rate increasing. This type of angina is defined as stable because there is a regular pattern for the pain and similar triggers exist for each attack of angina.

- Sufferers of stable angina can usually predict how an attack can happen, and these attacks usually only last for a couple of minutes.
  The pain is usually relieved by rest or by taking angina medication.

- **Unstable angina** Unstable angina does not follow a regular pattern. A small amount of physical effort such as walking, or even just resting, can result in an attack.
  In many cases, stable angina can progress to unstable angina, which may be an indication of worsening of coronary heart disease.

- In these cases, the patient should see a doctor urgently. Generally, pain from unstable angina is worse than with stable angina.
  The pain also tends to happen more often, and each episode may last as long as half an hour.

- **Variant angina** Variant angina is a rare form of angina.
  It is thought to occur as a result of coronary artery vasospasm, resulting in diminished blood flow.
  Variant angina is very painful and occurs from midnight to early morning.
  It usually occurs at rest and the same time each day.

Table 6-2 Classification and Treatment of Anginal Prototypes

Classification	Etiology	Symptoms	Treatment
Stable	Myocardial ischemia ACS	• Episodic pain lasting 5–10 minutes • Aggravated by exertion, exposure to cold, eating, emotional stress • Relieved by rest or nitroglycerin	• NTG • Aspirin • Beta adrenergic blockers • Calcium channel blockers • ACE inhibitors
Unstable	Ruptured or thickened plaque with platelet and fibrin thrombus, causing an increase in coronary obstruction	• Angina of increasing intensity, frequency, or duration • Occurs at rest or with minimal activity • Pain unresponsive to NTG	• NTG • Beta blockers tissue plasminogen activators (tPA), oxygen • Clopidogrel • Morphine Statins • Aspirin
Variant	Coronary vasospasm	• Occurs at rest without provocation • Triggered by smoking Transient ST-segment elevation during pain. Often with associated AV block or ventricular arrhythmias • Occurs with or without the presence of CAD	• Calcium channel blockers

Signs and symptoms

- The primary symptom of angina is chest pain, which is usually described as a ‘heaviness’ or a feeling of ‘pressure’ on the chest cavity. Severity of this pain varies from person to person, and patients can experience anything from a dull ache to severe pain.

- Sometimes angina pain may extend from the chest to the neck, down the arm and into the stomach, back, or jaw.
  In addition to the pain in these areas, other angina symptoms can include breathlessness, sweating and nausea.

- Angina symptoms develop when the heart demands more blood than is supplied to it by the coronary arteries.
  This increased demand for blood is usually brought on by physical activity, but can also include other triggers such as emotional upset, stress, cold weather or after a meal.

- Depending on the type of angina, symptoms usually last less than ten minutes once the person has stopped the activity that triggered the pain.

Management

- Healthcare professionals play a crucial role in the management of patients with angina.
  An accurate assessment of the patient must be carried out to ascertain the location, duration and the intensity of the pain.
  Risk factors must be identified to provide high-quality care.

- The care should include controlling pain, reducing anxiety and providing information on the possible risk factors and how to take appropriate measures to minimize the risk of developing angina.

- The main goals of treatment in angina pectoris are to relieve the symptoms, slow the progression of disease and reduce the possibility of future events, especially MI and premature death.

- Smoking cessation results in a significant reduction of acute adverse effects on the heart and may reverse, or at least slow, atherosclerosis.
  Strongly encourage patients to quit smoking, and take an active role in helping them to achieve this goal.

Pharmacological interventions

- A beta blocker or a calcium channel blocker is used as first-line treatment for stable angina. The choice of the drug to use is based on comorbidities, contraindications and the person's preference.

Coronary revascularization

- Coronary revascularization is required in those at high risk and those who fail to be controlled by medical therapy.
  A cardiac rehabilitation programme should be arranged following revascularization.

- Both coronary artery bypass grafting and percutaneous transluminal angioplasty have their indications and advocates. For the low-risk patient with stable angina, medical management carries the lowest risk.

Asthma

Lecture (20)

Overview of anatomy and physiology

- Asthma is caused by inflammation of the airways. These are the small tubes, called bronchi, which carry air in and out of the lungs.
  In asthma, the bronchi will be inflamed and more sensitive than normal.

- The airways of the lungs consist of the cartilaginous bronchi, membranous bronchi and gas-exchanging bronchi, termed the respiratory bronchioles and alveolar ducts (Figure 24.1).
  While the first two types function mostly as anatomic dead space, they also contribute to airway resistance.
  The smallest non-gas-exchanging airways, the terminal bronchioles, are approximately 0.5 mm in diameter; airways are considered small if they are less than 2 mm in diameter.

- Cellular elements include mast cells, which are involved in the complex control of releasing histamine and other mediators. Basophils, eosinophils, neutrophils and macrophages also are responsible for extensive mediator release in the early and late stages of bronchial asthma.

- Stretch and irritant receptors reside in the airways, as do cholinergic motor nerves, which innervate the smooth muscle and glandular units. In bronchial asthma, smooth muscle contraction in an airway is greater than that expected for its size if it were functioning normally, and this contraction varies in its distribution.

Pathophysiology

- Some of the principal cells identified in airway inflammation include mast cells, eosinophils, epithelial cells, macrophages and activated T lymphocytes.
  T lymphocytes play an important role in the regulation of airway inflammation through the release of numerous cytokines.

- Other constituent airway cells, such as fibroblasts, endothelial cells and epithelial cells, contribute to the chronicity of the disease.

  Other factors, such as adhesion molecules (e.g. selectins, integrins), are critical in directing the inflammatory changes in the airway.

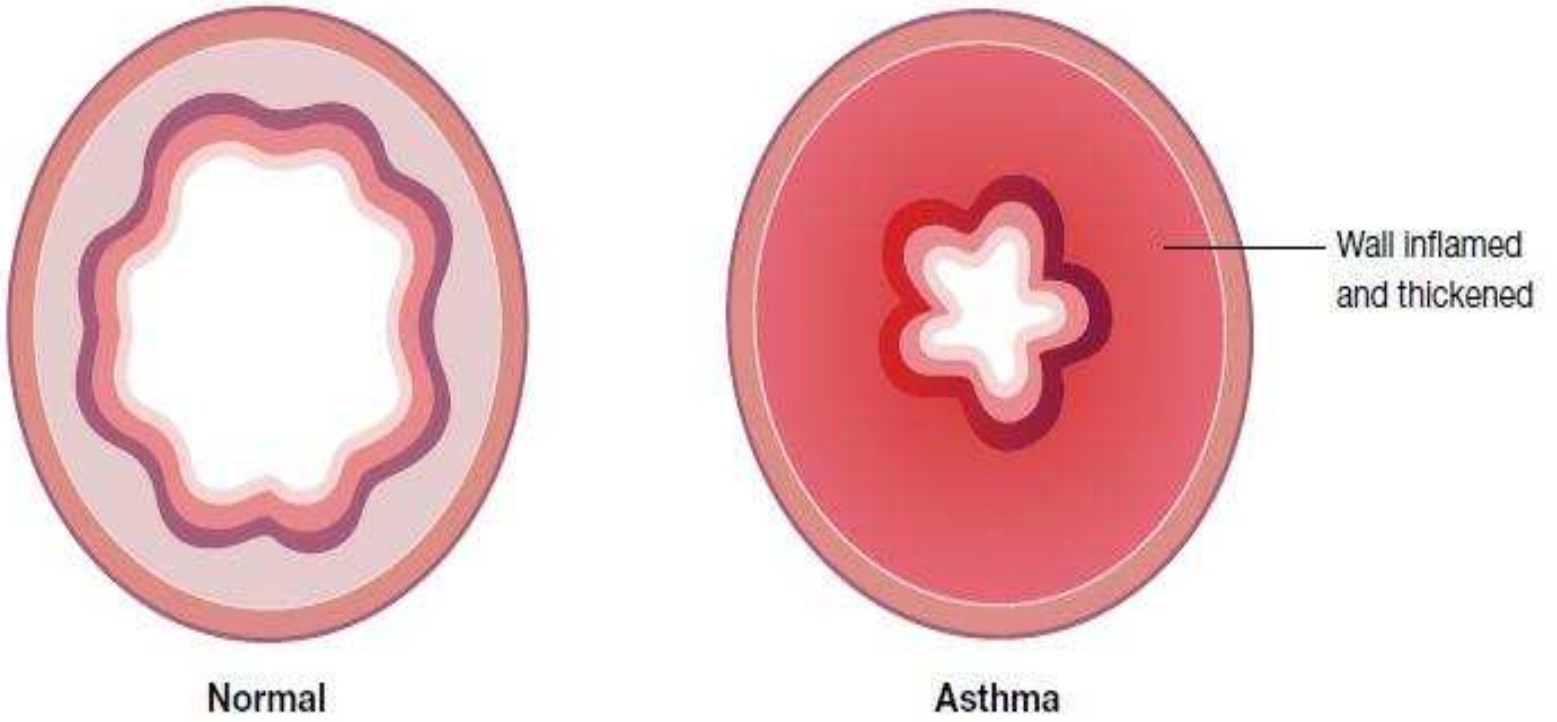
  Finally, cell-derived mediators influence smooth muscle tone and produce structural changes and remodelling of the airway.

- The presence of airway hyper-responsiveness or bronchial hyper-reactivity in asthma is an exaggerated response to numerous exogenous and endogenous stimuli. The mechanisms involved include direct stimulation of airway smooth muscle and indirect stimulation by

pharmacologically active substances from mediator-secreting cells such as mast cells or non-myelinated sensory neurons. The degree of airway hyper-responsiveness generally correlates with the clinical severity of asthma.

- Airway obstruction causes increased resistance to airflow and decreased expiratory flow rates.
  These changes lead to a decreased ability to expel air and may result in hyperinflation (Figure 24.2).
  The resulting over-distention helps maintain airway patency, thereby improving expiratory flow; however, it also alters pulmonary mechanics and increases the work of breathing.

Figure 24.2 Changes in the bronchioles in asthma



Bronchial hyper-responsiveness

- Hyperinflation compensates for the airflow obstruction, but this compensation is limited when the tidal volume approaches the volume of the pulmonary dead space; the result is alveolar hypoventilation.

- Uneven changes in airflow resistance, the resulting uneven distribution of air, and alterations in circulation from increased intra-alveolar pressure due to hyperinflation all lead to ventilation-perfusion mismatch. Vasoconstriction due to alveolar hypoxia also contributes to this mismatch. Vasoconstriction is also considered an adaptive response to ventilation/perfusion mismatch.

Signs and symptoms

- The symptoms of asthma can range from mild to severe. When asthma symptoms get significantly worse, it is known as an asthma attack.
  The symptoms include feeling breathless, a tight chest, like a band tightening around it, wheezing, which makes a whistling sound when the patient breathes, coughing, particularly at night,

- and early morning attacks triggered by exercise, exposure to allergens and other triggers.
  A severe asthma attack usually develops slowly, taking 6–48 hours to become serious.
  However, for some people, asthma symptoms can get worse quickly.

What causes asthma symptoms to occur?

- Many things can trigger or worsen asthma symptoms.
  A doctor will help a patient find out which things (called triggers) may cause asthma to flare up if there is contact with them. Triggers can include: allergens from dust, animal fur, cockroaches, mould, and pollens from trees, grasses and flowers,

- Irritants such as cigarette smoke, air pollution, chemicals or dust in the workplace, compounds in home decor products, sprays (such as hairspray), medicines (such as aspirin or other non-steroidal anti-inflammatory drugs and non-selective beta blockers), viral upper respiratory infections (such as colds) and chemicals in food and drinks.

Management

- **The aim of treatment** is to get asthma under control and keep it that way.
  Everyone with asthma should be able to lead a full and unrestricted life.
  The treatments available for asthma are effective in most people and should enable them to be free from symptoms.
  These should include: continuous monitoring of vital signs until the patient is stabilized, and safe administration of prescribed oxygen and medications.

- Regular PEFR (Peak Expiratory Flow Rate) measurement-singular or infrequent peak flows will not accurately reflect the patient's status.
  PEFR should be measured every 15–30 minutes after commencement of treatment until conditions stabilize.

- PEFR can also be used to measure the effectiveness of bronchodilator therapy; therefore, PEFR should be measured pre- and post-inhaled or nebulized beta-2 agonists at least four times a day throughout their stay in hospital.

- Health promotion – avoidance of triggers, compliance with prescribed pharmacological therapies, smoking cessation and weight reduction in obese patients may reduce the frequency of asthma attacks.

Chronic obstructive pulmonary disease (COPD)

Lecture (21)

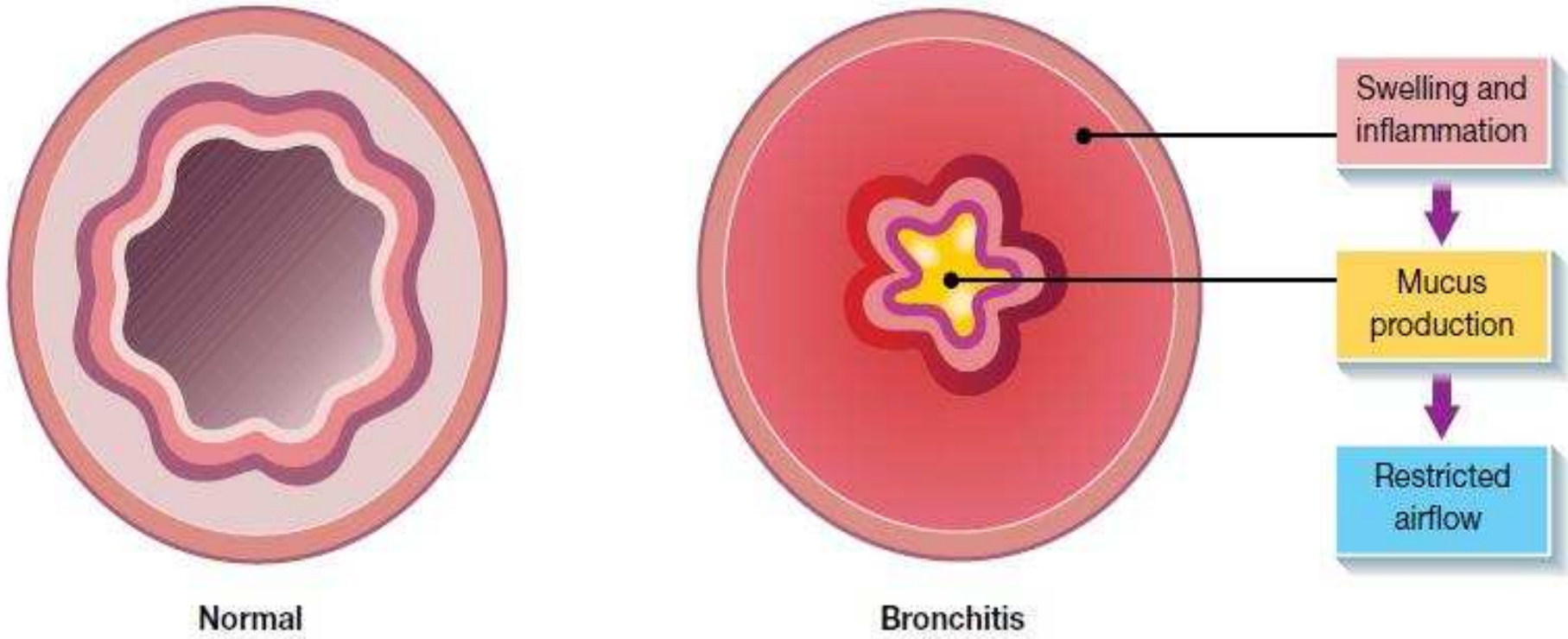
Overview of anatomy and physiology

- Chronic obstructive pulmonary disease (COPD) is the name for a collection of lung diseases including chronic bronchitis, emphysema and chronic obstructive airways disease.
  COPD affects middle-aged and older adults.

Pathophysiology

- COPD is characterized by slow progressive obstruction of the airways.
  COPD is a condition where the airways become inflamed and the air sacs in the lungs are damaged. This causes the airways to become narrower (Figure 27.1), which makes it harder to breathe in and out.

Figure 25.2 Narrowing of the inflamed bronchial tube



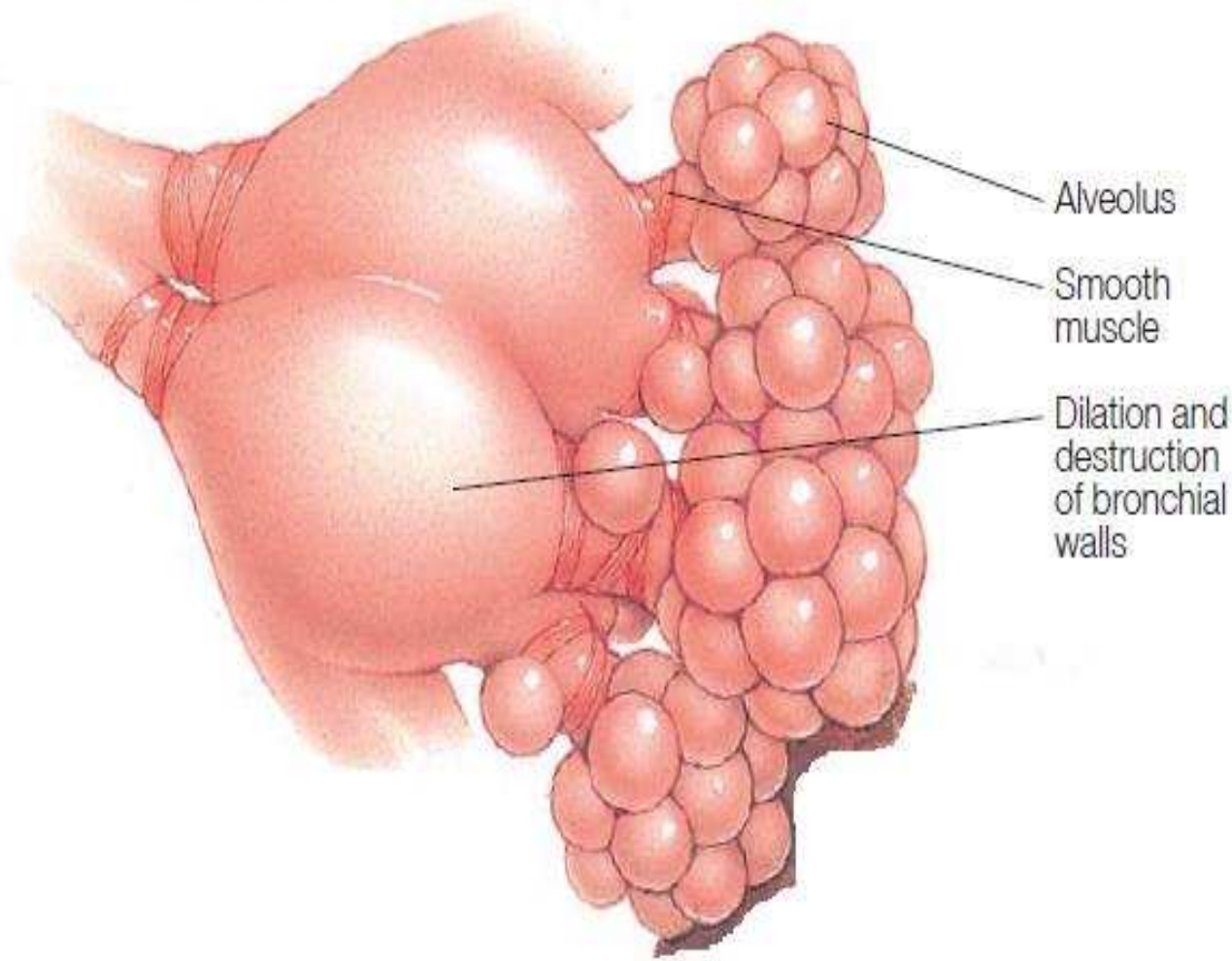
- The most consistent pathophysiological finding is hypertrophy and an increase in number of the mucus secreting goblet cells of the bronchial tree, evenly distributed throughout the lung but mainly seen in the larger bronchi.

- The small airways are particularly affected early in the disease, initially without the development of any significant breathlessness.
  This initial inflammation of the small airways is reversible and accounts for the improvement in airway function if smoking is stopped early.

- In later stages the inflammation continues even if smoking is stopped. Further progression of the disease leads to progressive squamous cell metaplasia and fibrosis of the bronchial walls.

- The physiological consequence of these changes is the development of airflow limitation.
  If the airway narrowing is combined with emphysema (causing loss of the elastic recoil of the lung with collapse of small airways during expiration) the resulting airflow limitation is even more severe.
  As the air sacs get permanently damaged, it will become increasingly difficult to breathe out.

Lung changes in emphysema



Causes of COPD

- Smoking is the main cause of COPD.
  At least four out of five people who develop the disease are, or have been, smokers.
  The lining of the airways becomes inflamed and permanently damaged by smoking.
  This damage cannot be reversed.
  Around 10–25% of smokers develop COPD.

- Exposure to certain types of dust and chemicals at work, including grains, isocyanates, cadmium and coal, has been linked to the development of COPD, even in people who do not smoke.
  Air pollution may be an additional risk factor for COPD.

- There is a rare genetic tendency to develop COPD, called alpha-1antitrypsin deficiency. This causes COPD in a small number of people (about 1%).
  Alpha-1-antitrypsin is a protein that protects the lungs. Without it, the lungs can be damaged by other enzymes that occur naturally in the body. People who have an alpha-1-antitrypsin deficiency usually develop COPD at a younger age, often under 35.

Signs and symptoms

The symptoms of COPD usually develop over a number of years, so the person may not be aware that they have the condition. The patient may have:

- Increasing breathlessness when exercising or moving around.
- A persistent cough with phlegm that never seems to go away.
- Frequent chest infections, particularly in winter.
- Wheezing.

- Symptoms of COPD are often worse in winter, and it is common to have two or more flare-ups a year.
  A flare-up (also known as an exacerbation) is when the symptoms are particularly bad. This is one of the most common reasons for people being admitted to hospital.

- Chest infections are more common if the person has COPD.
  Wheezing with cough and breathlessness may become worse than usual if the person has a chest infection and the person may cough more sputum. Sputum usually turns yellow or green during a chest infection.
  Chest infections can be caused by bacteria or viruses. Bacteria (which can be killed using antibiotics) cause about one in two or three exacerbations of COPD.

Viruses (which cannot be killed with antibiotics) are a common cause of exacerbations too, particularly in the winter months.
The common cold virus may be responsible for up to one in three exacerbations.
Other signs of COPD can include:

- Weight loss
- Tiredness and fatigue
- Swollen ankles

Management

- There is no cure for COPD, but treatment can help slow the progression of the condition and reduce the symptoms. The best way to prevent COPD from quickly getting worse is to stop smoking and avoid further damage to the lungs.

- There are also medicines that can help relieve the symptoms of COPD.
  The type of medicine prescribed will depend on how severe the COPD is and what symptoms the person has.
  Often, people with COPD have to take a combination of medicines.
  In addition, many people keep different medicines available in case they have a flare-up, when symptoms are particularly bad.

- The person may be prescribed short-acting and/or long-acting inhalers. There are two types of short-acting bronchodilator inhaler: beta-2 agonist inhalers, such as salbutamol and terbutaline and antimuscarinic inhalers, such as ipratropium.
  Long-acting bronchodilator inhalers: beta-2 agonist inhalers, such as salmeterol and formoterol and/or antimuscarinic inhalers, such as tiotropium, may be prescribed.

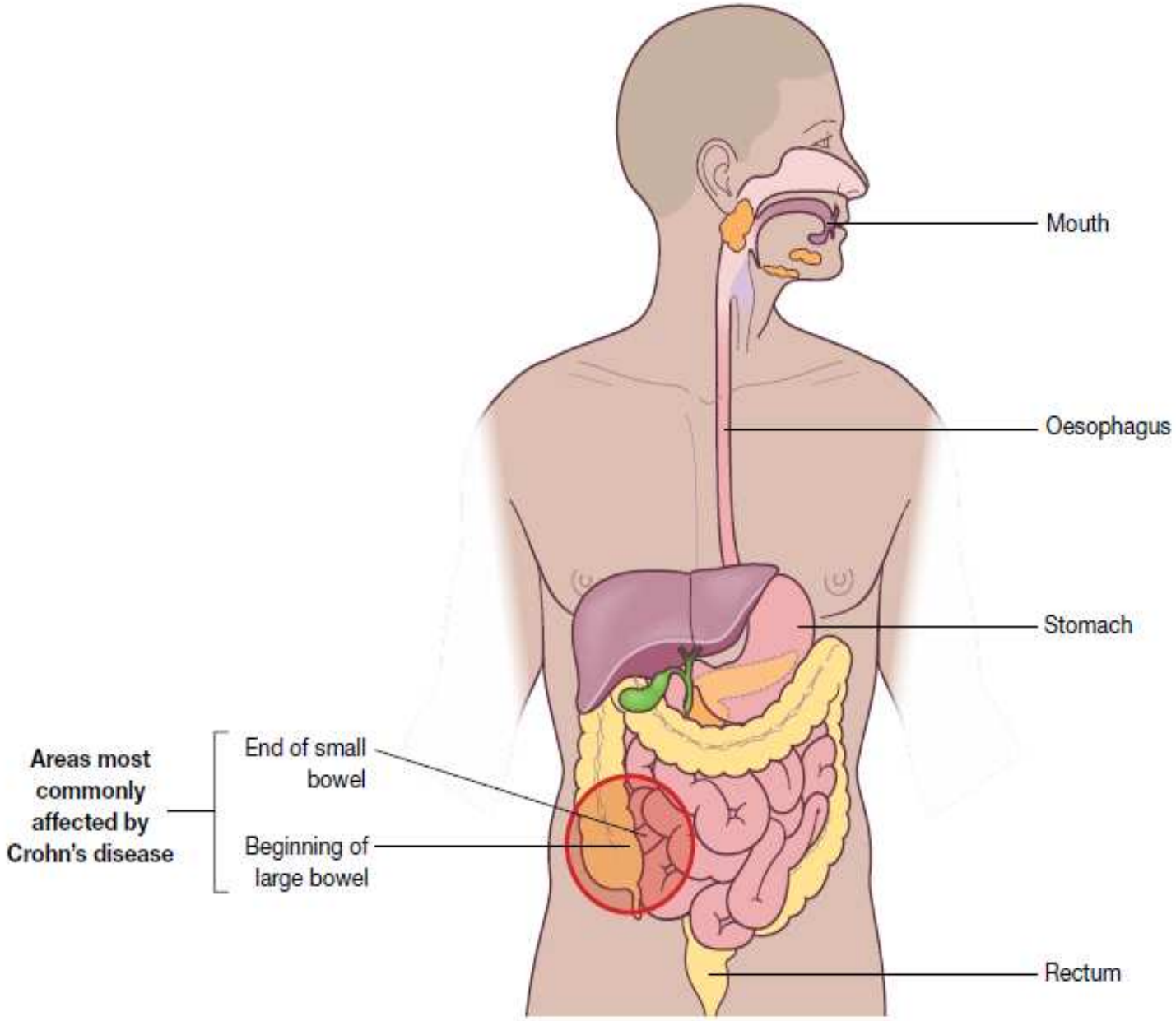
- Steroid inhalers, also called corticosteroid inhalers, work by reducing the inflammation in the airways.
- Other treatments include oxygen therapy and in some cases noninvasive ventilation (NIV) may be necessary to control breathing.

Crohn's disease

Overview of anatomy and physiology

- Crohn's disease is a long-term condition that causes inflammation of the lining of the digestive system. Any part of the gut can be affected (Figure 29.1).
  However, the most common site for the disease to start is the last part of the small intestine (the ileum).

Figure 29.1 Areas affected by Crohn's disease



- The gut (gastrointestinal tract) is the long tube that starts at the mouth and ends at the anus. When we eat, food passes down the esophagus , into the stomach and then into the small intestine.

The small intestine has three sections: the duodenum, jejunum and ileum. The small intestine is where food is digested and absorbed into the bloodstream. The structure of the gut then changes to become the large intestine (colon and rectum, sometimes called the large bowel).

The colon absorbs water, and contains food that has not been digested, such as fibre. This is passed into the last part of the large intestine, where it is stored as faeces. Faeces (stools) are then passed out of the anus into the toilet.

Pathophysiology

- A patch of inflammation may be small, or spread quite a distance along part of the gut.
  Several patches of inflammation may develop along the gut, with normal sections of gut in between.

- In about three in ten cases, the inflammation occurs just in the small intestine.
  In about two in ten cases the inflammation occurs just in the colon.
  In a number of cases, the inflammation occurs in different places in the gut.

- The lumen of the affected bowel assumes a ‘cobblestone’ appearance as fissures and ulcers surround islands of intact mucosa over oedematous submucosa (Figure 29.2).

- The inflammatory lesions of Crohn's disease are not continuous; rather, they often occur as 'skip' lesions, with intervening areas of normal-appearing bowel.
  Some evidence suggests that despite its normal appearance, the entire bowel is affected by this disorder.

- Depending on the severity and extent of the disease, malabsorption and malnutrition may develop as the ulcers prevent absorption of nutrients.
  When the jejunum and ileum are affected, the absorption of multiple nutrients may be impaired, including carbohydrates, proteins, fats, vitamins and folate.

- Disease in the terminal ileum can lead to vitamin B12 malabsorption and bile salt reabsorption.
  The ulcerations can also lead to protein loss and chronic, slow blood loss, with consequent anaemia.

Risk factors

The exact cause of Crohn's disease is unknown. However, some possible factors include: some genetic factors; the inflammation may be caused by a problem with the immune system (the body's defence against infection and illness) that causes it to attack healthy bacteria in the gut;
smokers with Crohn's disease usually have more severe symptoms than non-smokers;

- Previous infection may trigger an abnormal response from the immune system;
  and Crohn's disease is most common in westernized countries, such as the UK, and least common in poorer parts of the world, such as Africa.

Signs and symptoms

The symptoms of Crohn's vary depending on which part of the digestive system is inflamed. Some of the common symptoms include diarrhoea, abdominal pain, fatigue, weight loss, fever, malaise and anaemia. Stools are liquid or semi-formed and do not contain blood. There may be long periods of mild symptoms followed by periods where the symptoms are particularly troublesome.

Management

- **Diet**

Eating a healthy, balanced diet is important for everyone, including in Crohn's disease. Patients may find that they are sensitive to certain foods which can make the symptoms worse. If this happens, then removing these foods from the diet can help.

- Always get advice from a dietitian before making any changes to the diet.

  During a flare-up, a liquid diet made up of simple forms of protein, carbohydrates and fats may help to ease the symptoms.

Medicines

- Some of the medication used in the treatment of Crohn's disease includes corticosteroids (such as hydrocortisone, beclometasone and budesonide). These are very effective but are usually used for a short time during a flare-up because they can cause weight gain and side effects such as diabetes and osteoporosis.

- Immunosuppressants may also be used to suppress the immune system (such as azathioprine, mercaptopurine and methotrexate), and antibiotics to reduce the risk of infection.

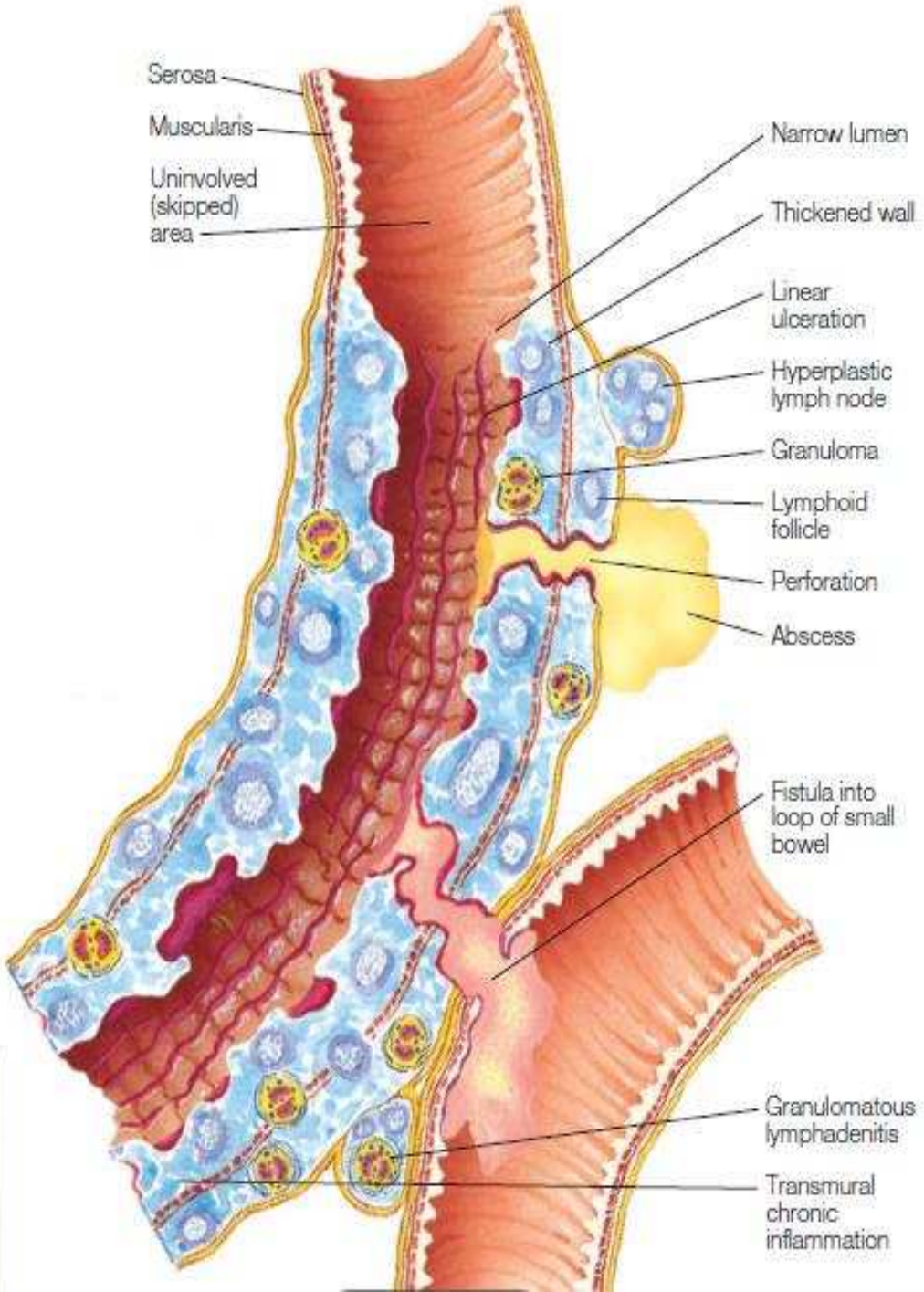
Surgery

- Surgery is often required when the symptoms of Crohn's disease cannot be controlled using medication alone.
- An estimated 80% of people with Crohn's disease do require surgery at some point in their life.

- Surgery cannot cure Crohn's disease but it can provide long periods of remission, often lasting several years. During surgery, the inflamed section of the digestive system is removed and the remaining part is reattached.

Complications

- Certain complications of Crohn's disease (e.g. intestinal obstruction, abscess and fistula) are so common that they are considered part of the disease process.
  For many clients, the disease initially presents with one of these complications.


Serosa
Muscularis
Uninvolved (skipped) area
Narrow lumen
Thickened wall
Linear ulceration
Hyperplastic lymph node
Granuloma
Lymphoid follicle
Perforation
Abscess
Fistula into loop of small bowel
Granulomatous lymphadenitis
Transmural chronic inflammation

- Intestinal obstruction is a common complication caused by repeated inflammation and scarring of the bowel that leads to fibrosis and stricture. Obstruction of the bowel lumen causes abdominal distention, cramping pain, and nausea and vomiting.

- Perforation of the bowel is uncommon, but can lead to generalized peritonitis.
- Massive haemorrhage also is an uncommon complication of Crohn's disease.
- Long-standing Crohn's disease increases the risk of cancer of the small intestine or colon by 5–6 times.

Peptic Ulcer

Overview of anatomy and physiology

- Food passes down the oesophagus into the stomach (Figure 30.1).
  The stomach makes acid which is not essential, but helps to digest food.
  After being mixed in the stomach, food passes into the duodenum (the first part of the small intestine).

- In the duodenum and the rest of the small intestine, food mixes with enzymes (chemicals).
  The enzymes come from the pancreas and from cells lining the intestine.
  The enzymes break down (digest) the food which is absorbed into the body.

- The stomach may be divided into seven major sections.
  The cardia is a 1–2 cm segment distal to the oesophagogastric junction.
  The fundus refers to the superior portion of the stomach that lies above an imaginary horizontal plane that passes through the oesophagogastric junction.

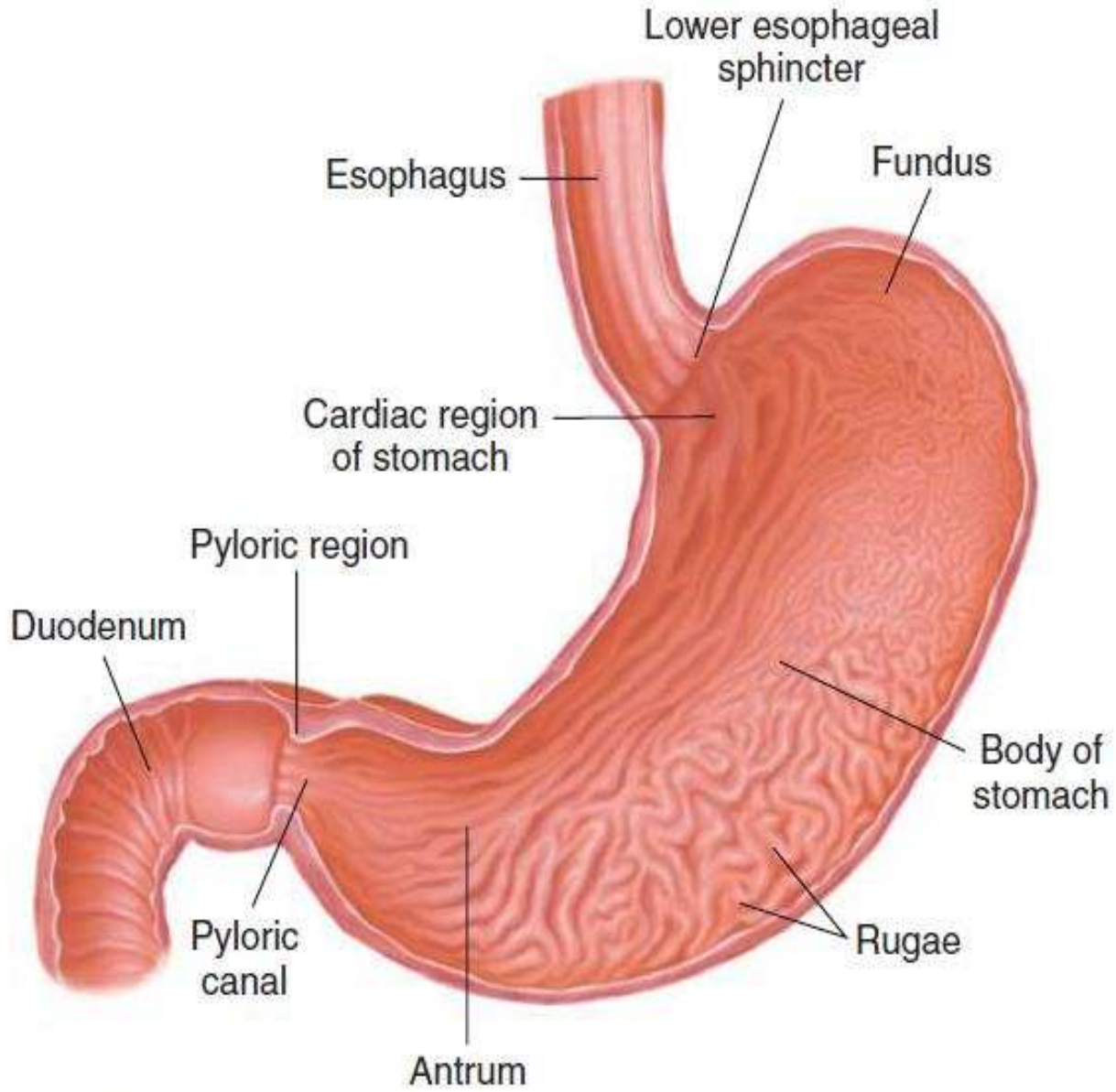


FIGURE 36-2 • Structures of the stomach.

- The antrum is the smaller distal a quarter to a third of the stomach.
  The narrow 1–2 cm channel that connects the stomach and duodenum is the pylorus.
  The lesser curve refers to the medial shorter border of the stomach, whereas the opposite surface is the greater curve.

Pathophysiology

- Ulcers are caused when there is an imbalance between the digestive juices produced by the stomach and the various factors that protect the lining of the stomach.
  Symptoms of ulcers may include bleeding.
  On rare occasions, an ulcer may completely erode the stomach wall.

- Mucus lines the digestive tract and acts as a barrier against the acidic gastric secretions. Too little mucus production coupled with too much acid production will leave the digestive tract vulnerable to acid erosion and ulceration. Erosion of the mucosal lining may result in the formation of a fistula. The fistula would allow the acidic gastric contents to leak out into the peritoneum, resulting in peritonitis.

- Stress, caffeine, cigarette smoking and alcohol consumption increase acid production.
  Medications such as NSAIDs and aspirin inhibit prostaglandins which protects mucosal lining.

Causes

- A peptic ulcer is an area of damage to the inner lining (the mucosa) of the stomach or the upper part of the intestine (duodenum) (Figure 30.2). A bacterium, *Helicobacter pylori*, is the main cause of ulcers in this area.
  H. pylori bacterial (Figure 30.3) infection leads to death of the mucosal epithelial cells of the stomach and duodenum.

- The bacteria release toxins and enzymes that reduce the efficiency of mucous in protecting the mucosal lining of the gastrointestinal tract.
  In response to the bacterial infection, the body initiates an inflammatory response which results in further destruction of the mucosal lining and ulceration.

- However, the ulcer can be caused by the use of painkillers called non-steroidal anti-inflammatory drugs (NSAIDs), such as aspirin, naproxen (Aleve, Anaprox, Naprosyn and others), ibuprofen (Motrin, Advil, Midol and others), and many others available by prescription.
  Even safety-coated aspirin and aspirin in powdered form can frequently cause ulcers.

- Other causes include excess acid production from gastrinomas, tumours of the acid producing cells of the stomach that increase acid output (seen in Zollinger-Ellison syndrome).
  Smoking, stress, excessive consumption of caffeine and familial history have also been included as risk factors for peptic ulcer.

Signs and symptoms

- Some patients with peptic ulcer have no symptoms. However, for some the possible symptoms include epigastric pain, heart burn, poor appetite, burping and feeling bloated.
  This pain is often described as burning or gnawing and may extend to the back.
  It usually comes on after eating – often 1–2 hours after a meal – and may come and go for several days or weeks.

Management

- Once peptic ulcer has been diagnosed, the nurse should help the patient identify any lifestyle factors which may be associated with peptic ulcers, such as stress, heavy alcohol consumption, smoking or drinking a lot of coffee.
  Once identified, the nurse and patient can discuss ways of reducing the risks.

- Nearly all duodenal ulcers are caused by infection with *H. pylori*.
  Therefore, a main part of the treatment is to clear this infection.
  If this infection is not cleared, the ulcer is likely to return once acid suppressing medication is stopped.

- Two antibiotics are needed.
  In addition, an acid-suppressing medicine is needed to reduce the acid in the stomach, this is needed to allow the antibiotics to work well.
  The patient may need to take this combination therapy (sometimes called triple therapy) for a week.

Medication

- A 4–8 week course of a medicine that greatly reduces the amount of acid that the stomach makes is usually advised. The most commonly used medicine is a proton pump inhibitor (PPI).
  This is a class (group) of medicines that work on the cells that line the stomach, reducing the production of acid. **They include:**
- esomeprazole, lansoprazole, omeprazole, pantoprazole and rabeprazole.

- Sometimes another class of medicines called H2 blockers is used. They are also called histamine H2-receptor antagonists, but are commonly called H2 blockers.
  H2 blockers work in a different way on the cells that line the stomach, reducing the production of acid.
  They include: cimetidine, famotidine, nizatidine and ranitidine, and come in various brand names.
  As the amount of acid is greatly reduced, the ulcer usually heals.

Diet

- Dietary advice should be offered to the patient. Small regular meals are encouraged, approximately five small meals per day to prevent hunger pain.
  Spicy food should be avoided as it may cause irritation on the mucosal membrane of the stomach, resulting in inflammation and epigastric pain.

Renal Failure

Overview of anatomy and physiology

- There are two kidneys, which are located in the abdominal cavity and lie at a slightly oblique angle on either side of the vertebral column at the levels of T12 through L3.
  They are approximately 10–12 cm long, 5–7 cm wide and 3–4 cm thick. They are beanshaped organs, where the outer surface of the kidney is convex and the inner surface is concave in shape.

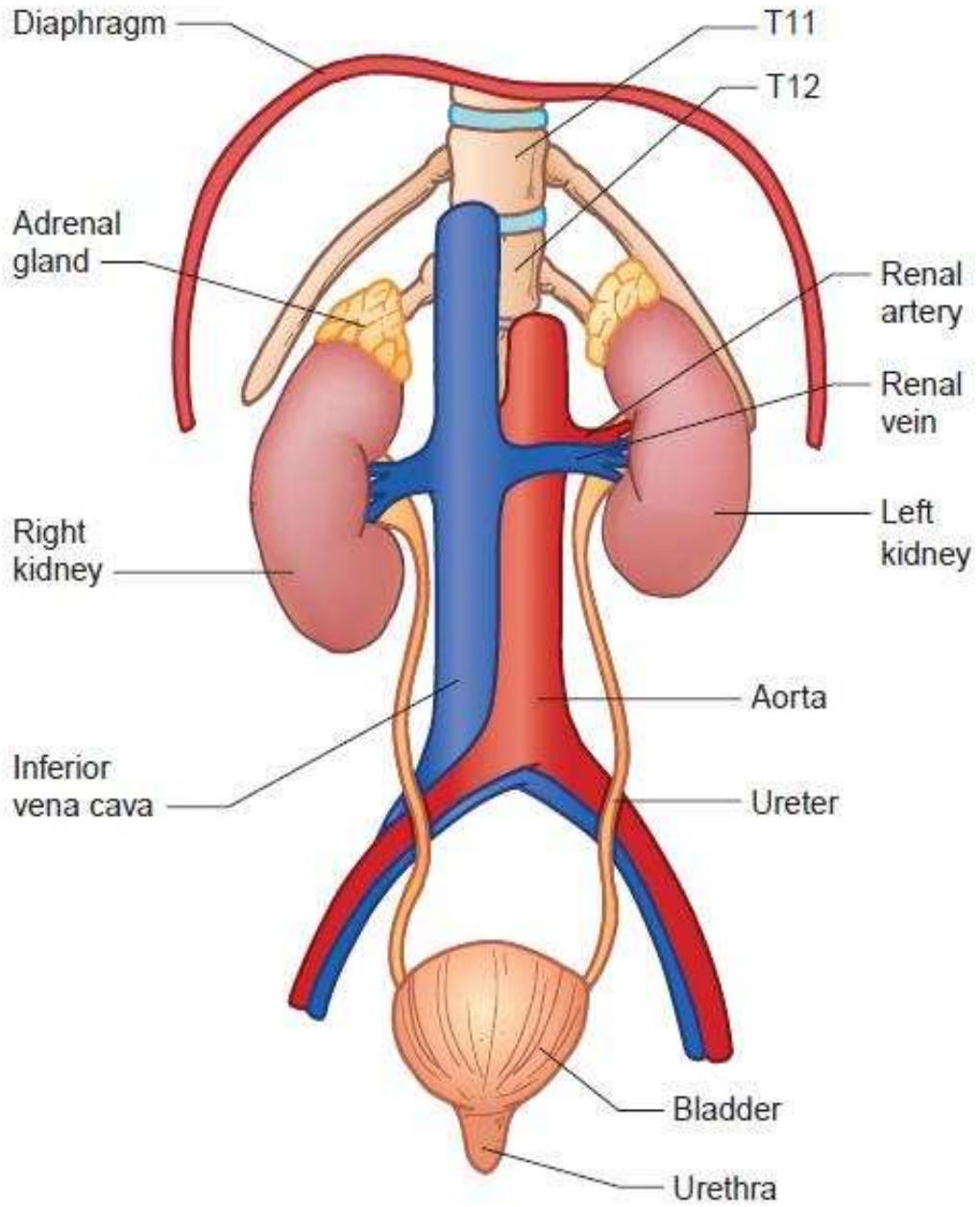


FIGURE 38.1 • Kidneys, ureters, and bladder. (The right kidney is usually lower than the left.)

Overview of anatomy and physiology

- Near the centre of the concave border is an indentation called renal hilum through which the ureter, renal artery, renal vein, lymphatic vessels and nerves enter and exit the kidney (Figure 37.1).

- The nephron is the structural and functional unit of the kidney. Each nephron functions as an independent unit and produces a miniscule quantity of urine.
  The nephron can be differentiated into the following regions: Bowman’s capsule, glomerulus, proximal convoluted tubule, loop of Henle, distal convoluted tubule and collecting ducts (Figure 37.2).

Blood supply

- The kidneys receive their blood supply directly from the aorta via the renal arteries and blood is returned to the inferior vena cava via the renal veins.
  The kidneys receive approximately 20% of the cardiac output.
  The blood supply to the kidneys arises from the paired renal arteries at the level of L2.
  They enter into the renal hilum, the passageway into the kidney, with the renal vein anteriorly; the renal artery and the renal pelvis posteriorly.

Blood supply

- The renal veins return the blood from the kidneys to the renal artery at the hilum. The left renal vein is longer than the right as it crosses the midline to reach the inferior vena cava (IVC).

Functions

- The kidneys serve important functions, including filtration and excretion of metabolic waste products (urea and ammonium); regulation of necessary electrolytes, fluid and acid-base balance; and stimulation of red blood cell production.

Functions

- They also serve to regulate blood pressure via the renin-angiotensin-aldosterone system, controlling reabsorption of water and maintaining intravascular volume.
  The kidneys also reabsorb glucose and amino acids and have hormonal functions via erythropoietin, calcitriol and vitamin D activation.

Pathophysiology
Acute kidney injury

- AKI is a rapid deterioration of renal function, resulting in inability to maintain fluid, electrolyte and acid-base balance.
  It is detected and monitored by serial serum creatinine readings primarily, which rise acutely, and urine output and GFR will decrease.

Aetiology

- **Prerenal:** These include volume depletion (e.g. haemorrhage, severe vomiting or diarrhoea, burns, inappropriate diuresis); cardiac failure; cirrhosis; nephrotic syndrome; hypotension. **Renal hypoperfusion**: these include non-steroidal anti-inflammatory drugs (NSAIDs) or selective cyclo-oxygenase-2 (COX-2) inhibitors; angiotensin-converting enzyme (ACE) inhibitors or angiotensin-II receptor antagonists (AIIRAs – commonly called angiotensin receptor blockers (ARBs)); abdominal aortic aneurysm; renal artery stenosis or occlusion; hepatorenal syndrome.

- **Intrarenal Glomerular disease**: this includes glomerulonephritis; thrombosis; haemolytic uraemic syndrome.
  Tubular injury: this includes acute tubular necrosis (ATN) following prolonged ischaemia; nephrotoxins (e.g. aminoglycosides, radiocontrast media, myoglobin).
  Vascular disease: this includes vasculitis (usually associated with antineutrophil cytoplasmic antibody); cryoglobulinaemia; polyarteritis nodosa; thrombotic microangiopathy; cholesterol emboli; renal artery stenosis; renal vein thrombosis; malignant hypertension; eclampsia.

- **Postrenal:** This includes calculus; prostatic hypertrophy or malignancy; retroperitoneal fibrosis; bladder tumours; pelvic malignancy; urethral strictures; papillary necrosis.

Chronic kidney disease

- Any condition that destroys the renal function can lead to CKD. The pathophysiology of CKD involves a gradual loss of entire nephron units.
  Chronic kidney disease is divided into five stages of increasing severity.
  In the early stages, as nephrons are slowly destroyed, remaining healthy nephrons take over the functions but become hypertrophied.

Chronic kidney disease

- The blood pressure and the filtration in the remaining healthy nephrons increase to compensate for the loss of nephrons, resulting in damage of the remaining healthy nephrons. This process of persistent loss of nephron function may continue even after the initial disease process has resolved.

Signs and symptoms

- Symptoms can vary from person to person.
  Someone in early stage kidney disease may not feel sick or notice symptoms as they occur.
  When kidneys fail to filter properly, waste accumulates in the blood and the body, a condition called azotaemia.
  Very low levels of azotaemia may produce few, if any, symptoms.
  Renal failure accompanied by noticeable symptoms is termed uraemia.

- Other symptoms include: weight loss; blood in the urine; vomiting and diarrhoea; swelling of the legs, ankles, feet, face and/or hands; shortness of breath due to extra fluid on the lungs; difficulty in micturition.

Management

- The management of renal failure depends which type it is.
  A full nursing assessment of vital signs, weight, fluid intake and output, nursing history and assessment of the patient's knowledge of the disease process should all be carried out in order to provide high quality care.

- The primary focus in drug management for AKI is to restore and maintain renal perfusion and to eliminate drugs that are nephrotoxic from the treatment regimen. Adequate hydration is of paramount importance to the patient’s management.

- Strict nutritional status should be maintained.

Protein intake should be limited to minimize the increase of nitrogenous wastes.
Carbohydrates are increased to maintain adequate calorie intake and provide a protein-sparing effect.
Where necessary, the dietician may be involved in the care of the patient with AKI.

- In CKD, unlike carbohydrates and fats, the body is unable to store excess proteins.
  Any unused protein is broken down as urea and nitrogenous waste, which are normally excreted by the kidneys.
  In CKD, these waste products are retained by the body, resulting in the toxic build-up of these waste products producing uraemic symptoms.
  However, prolonged dietary protein restriction should be avoided. Once the dialysis has commenced, a high protein diet is recommended.

- Water and sodium intake are regulated to maintain the extracellular fluid volume at normal levels.
  Strict water and sodium restrictions may be necessary as the CKD progresses.

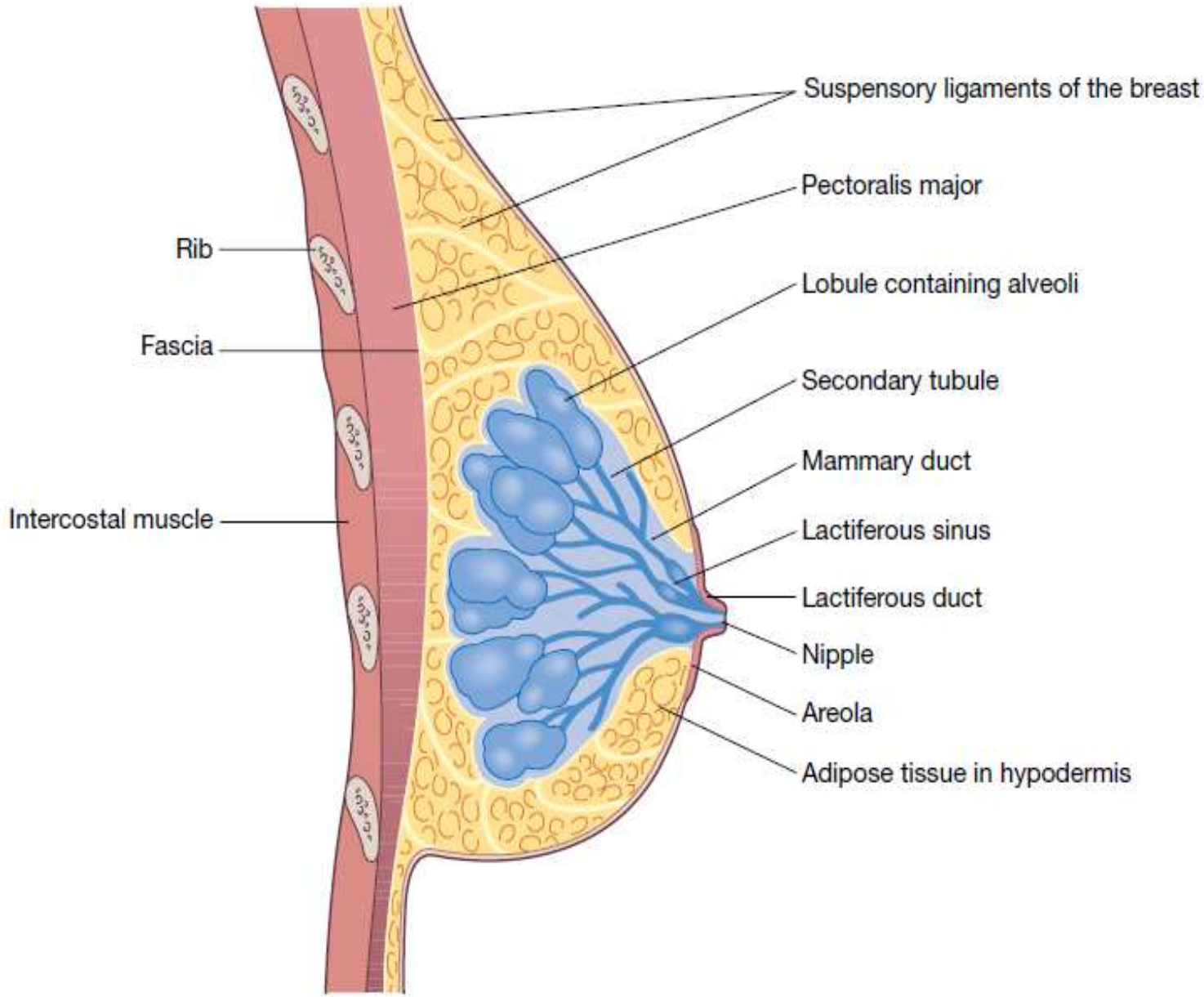
Cancer breast

- Globally, breast cancer is the most frequently diagnosed life threatening cancer in women and is the leading cause of cancer death among women. Breast cancer can be discovered while in situ (localized) or as a malignant neoplasm (spreading).

The female breasts

- An adult woman's breasts are milk-producing glands on the front of the chest wall, resting on the pectoralis major, supported by and attached to the front of the chest wall on either side of the sternum by ligaments.
  There are 15–20 lobes arranged in a circular fashion in each breast. Each lobe comprises many lobules, with glands that produce milk in response to hormones (see Figure 48.1).

Figure 48.1 The female breast



- A number of early breast carcinomas are asymptomatic, particularly if they are discovered during a breast-screening programme.
  Larger tumours can present as a painless mass.
  Pain or discomfort is not usually a symptom of breast cancer.
  A large number of breast cancer cases in the UK can be explained by factors that influence exposure to oestrogen, including reproductive and hormonal factors, obesity, alcohol and physical activity.

- Breast cancers are predominantly linked to modifiable lifestyle and environmental factors (see Figure 48.2).

Figure 48.2 Risk factors

- Age (increases with age)
- Reproductive history
- Endogenous and exogenous hormones
- Breast density
- Previous breast disease
  - the genes BRCA1, BRCA2 and TP53 mutations carry very high risk. Family history, a first-degree relative, is the most widely recognized breast cancer risk factor
- Non-reproductive lifestyle

Pathophysiology

- The majority of breast cancers arise from either the epithelial lining of ducts, known as ductal, or from the epithelium of the terminal ducts of the lobules and these are called lobular.
  Carcinoma can be invasive or in situ.
  Most cancers arise from intermediate ducts and are invasive. They eventually grow through the wall of the duct and into the fatty tissue.

- Current understanding of breast cancer is that invasive cancers arise through a series of molecular alterations at the cell level.
  These alterations result in breast epithelial cells with remarkable features and uncontrolled growth.

- An infiltrating carcinoma of the nipple epithelium, called Paget's disease of the breast, accounts for approximately 1% of all breast cancers.
  Inflammatory carcinoma exists in less than 3% of all cases; there is a rapidly growing, sometimes painful mass increasing the breast and causing the skin covering it to become red and warm. There may be widespread infiltration of tumour.

Staging of the tumour

- Tumour size, lymph node status and distant metastasis are taken into account when staging the tumour (see Table 48.1).
  The various tests and investigations required to make a diagnosis can also give some information about the stage.
  Understanding the stage is important as this helps the clinician to decide on the most appropriate treatment.

Table 48.1 Breast cancer staging

Stage 0	Carcinoma *in situ* and is not invasive
Stage I	Tumour up to 2 cm in diameter and no lymph node involvement or metastasis
Stage II	Tumour 2–5 cm in diameter or there is spread to the axillary lymph nodes on the same side and the nodes are not adherent
Stage IIIa	The tumour is over 5 cm in diameter or the nodes are adherent
Stage IIIb	Invasive breast cancer in which a tumour of any size has spread to the breast skin, chest wall or internal mammary lymph nodes and includes inflammatory breast cancer with peau d'orange
Stage IV	Is spread beyond the breast, axilla and internal mammary nodes. It may have spread to supraclavicular nodes, bone, liver, lung or brain

Signs and symptoms

- Most early breast carcinomas are asymptomatic, especially if discovered during a breast-screening programme.
  Larger tumours may present as a painless mass.
  Pain (mastalgia) or discomfort is not usually a symptom of breast cancer; only 5% of women with a malignant mass have breast pain.

- Most women will have felt a lump, there may be nipple change and in some cases nipple discharge; skin contour changes can occur.
  Intraduct carcinoma can present as a bloody discharge from the nipple.

Investigations

- A variety of investigations are used; however, evaluation starts with an ordered inquiry that begins with symptoms and a general clinical history, along with a physical examination of the breasts, the axillary and cervical lymph nodes. Once the examination has been concluded other investigations may be required; these are used to make a diagnosis and to help with staging.

- Investigations should be aligned to local national guidance.
  Triple assessment includes the following components:
  - clinical examination,
  - imaging (including mammography, ultrasonography or both)
  - and needle biopsy.

- Mammography is more appropriate for less dense breasts and is almost always undertaken; ultrasound and mammography can detect more invasive tumours
- Ultrasound is effective particularly when breast tissue is dense; in the younger woman it can be diagnostically more useful than mammography.

- MRI is used in women with dense breast tissue, cases of familial breast cancer associated with BRCA mutations, silicone gel implants, positive axillary lymph node status with occult primary tumour in the breast or if multiple tumour foci are suspected.

- Core needle biopsy (image-guided) should be obtained before any surgery. Ultrasound or stereotactic mammographic guidance can be used.
- Open biopsy (needle localization), radio-opaque needles are used to guide biopsy (this can be done under local anaesthetic).

- Where there are palpable lesions fine-needle aspiration is required.
- Core needle biopsy is usually used for larger lesions.
- Excision biopsy (under local anaesthesia) allows the entire lesion to be removed).
- Incisional biopsy provides for part of a lesion to be removed (lesions 4 cm or larger).

- Specific staging investigations include routine blood tests, such as liver function tests, chest X-ray; if metastases are suspected a CT scan is needed.
  Bone scintigraphy (Bone scan) is required if there are distant metastases.
  Positron emission tomography (PET) can detect distant metastases.

Management

- Treatment must be patient-centred, acknowledging the woman's individual needs and preferences.
  Discussion and involvement of the woman's family should, with their consent, be enabled.
  Prognosis of patients with breast cancer depends on biological characteristics of the cancer, the patient and on appropriate therapy.

- The primary treatment for early stage breast cancer is surgery; many women are cured with surgery alone.
  The aim of breast cancer surgery includes complete resection of the primary tumour with negative margins, reducing the risk of local recurrences, and pathological staging of the tumour and axillary lymph nodes providing necessary prognostic information.

- Adjuvant treatment of breast cancer is intended to treat micrometastatic disease (breast cancer cells escaping into the breast and regional lymph nodes but which have not yet had an established identifiable metastasis).
  Adjuvant treatment involves radiation therapy and a number of chemotherapeutic and biological agents.

- Treatment attempts to reduce the risk of future recurrence, and in so doing reduce breast cancer-related morbidity and mortality.